Fuzzy PI Control for bittide: Nonlinear Analysis and Transient Performance

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Abstract

Finite receiving buffers couple synchronization performance to transient resource use in bittide networks. We propose a decentralized incremental Fuzzy PI controller that shapes oscillator corrections using occupancy error and its update-to-update change. We specify its full inference law, derive an implementation-consistent PI calibration and characterize its central nonlinear map. For a synchronous zero-delay fluid reference, an invariant-domain and Lyapunov analysis, combined with interval enclosures and a contracting tail, certifies an all-time buffer-excursion benefit for a prescribed initialization: below 17.315 frames for Fuzzy versus approximately 23.404 for matched-gain PI. Continuous-phase Callisto experiments separately evaluate delayed local execution. After equal-budget selection from 24 candidates per family, 186 untouched tests show mean paired disagreement-RMS improvements of 16.78% and 7.85% in two bounded-random latency regimes. Matched-gain startup tests also show smaller buffer excursions. These gains depend on the operating regime and carry a common-frequency-shift cost; slower-random confirmation delimits their applicability. The results connect an explicit nonlinear mechanism to conditional headroom and synchronization benefits.

CCS Concepts

• **Computing methodologies**;

Keywords

Distributed synchronization, bittide, fuzzy control, logical synchrony, cyber-physical systems

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1 Introduction

Distributed cyber-physical computation requires nodes running on independent clocks to coordinate their computations and communications. For applications that depend on reproducible execution, the relationships between communicated events must remain well defined as physical timing varies. This creates a fundamental implementation challenge: a coherent logical execution must be realized using oscillators, communication links and storage resources whose physical behaviour is neither identical nor unlimited.

Logical synchrony provides a framework for addressing this challenge without distributing a global physical clock. In bittide, continuous frame exchange and elastic receiving buffers connect the physical operation of neighbouring nodes to their logical execution. Each node adjusts its oscillator frequency using locally observed buffer occupancies. These buffers consequently serve two roles: they provide the feedback signal for synchronization and absorb the imbalance between incoming frames and local consumption. The storage available to accommodate that imbalance is finite [3].

The resulting control problem concerns the entire trajectory towards an operating point. Initial oscillator differences and changes in frame-delivery time can consume buffer headroom before the network settles or while it responds to a disturbance. A small eventual frequency disagreement does not by itself establish that the intervening buffer trajectory remains admissible. The objective is therefore to use local occupancy feedback to reduce frequency disagreement and control deviations from compatible occupancy targets, while respecting finite buffer capacity and oscillator-correction limits. This formulation makes transient buffer excursion, convergence and control activity joint criteria for evaluating a controller.

Existing bittide research establishes important foundations for this problem. PI performance analysis [1] relates closed-loop behaviour to controller gains and network resistance distances. Fluid occupancy analysis [4] characterizes steady buffer levels, and reframing methods [2] address buffer centering. Our earlier study, *Stabilizing the bittide by Tuning Controllers Under Topology and Latency Constraints*, examined PI gain selection under topology and fixed-delay constraints. Building on these results, the question addressed here is how a local nonlinear Fuzzy PI law can reshape transient resource use under comparable scales, memory and update conventions. In particular, the connection between the nonlinear rule surface, a provable buffer-excursion benefit and behaviour under delivery perturbations requires explicit analysis and controlled comparison.

We propose a decentralized occupancy-based Fuzzy PI controller for bittide. Each node computes an oscillator-correction increment from its receiving-buffer error and the update-to-update change in that error. In the central input region studied here, the proposed surface strengthens its calibrated linear counterpart for same-sign inputs and attenuates its magnitude for opposing-sign inputs. This direction-dependent action provides flexibility at fixed calibrated PI gain coordinates. It motivates investigating whether the nonlinear correction can reduce transient buffer demands while retaining convergence, and determining how its effects on frequency disagreement and control activity depend on the operating conditions.

This paper presents the complete controller, establishes its continuous-clock realization and combines nonlinear analysis with evaluation to investigate that question. We derive the exact central-region map and its scale correspondence with native PI, and verify implementation equivalence when the nonlinear map is replaced by its calibrated linear counterpart. For a synchronous, zero-delay

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fluid reference, we establish an invariant input region and a common Lyapunov certificate. An outward interval enclosure of the nonlinear trajectory, followed by a contracting infinite-tail bound, provides an all-time buffer-excursion guarantee. For one prescribed three-node initialization at matched gain coordinates, the Fuzzy excursion is certified below 17.315 fluid frames, whereas PI attains an exact peak of approximately 23.403701 frames. This establishes a strict buffer-headroom benefit for that reference and initialization.

We separately evaluate the proposed law in continuous-phase Callisto, including integer buffer events, local-clock polling and delayed correction application. Matched-gain startup tests show smaller buffer excursions for Fuzzy at three prescribed initial frequency mismatches. We then give PI and fixed-shape Fuzzy the same 24-candidate, four-case tuning budget, freeze the selected controllers, and execute 186 untouched tests using shared delivery profiles. In the two original bounded-random regimes, the random-priority Fuzzy controller reduces mean paired disagreement-RMS ratios by 16.78% and 7.85% relative to the independently selected PI, with lower mean command activity. The burst-priority Fuzzy controller also reduces IAE by approximately 55.8% in the larger held-out burst. These gains have a defined operating scope: common-frequency shift is a cost, and a smaller burst and a separately registered slower-random regime do not share the improvement. The combined evidence establishes conditional transient and synchronization benefits while keeping the reference guarantee distinct from measured delayed execution.

The contributions of this paper are:

(1) A decentralized incremental Fuzzy PI controller for bittide, with complete inference rules, continuous-clock realization, implementation-checked PI calibration and an exact central-region nonlinear characterization.
(2) An invariant-domain and Lyapunov analysis for the stated fluid reference, with bounds on physical frequencies and applied corrections, and a strict all-time reduction in buffer excursion for one prescribed initialization at matched gains.
(3) Continuous-phase Callisto evaluation combining matched-gain startup stress, equal-budget independent tuning and untouched paired tests, identifying operating conditions with lower disagreement RMS or transient buffer demand and their associated performance tradeoffs.

2 Background

2.1 Logical Synchrony and the bittide Mechanism

Logical synchrony describes relationships between distributed events without requiring a global physical clock. In bittide, neighbouring nodes exchange ordered frames, and elastic receiving buffers accommodate differences in their physical progress. The association between sending and receiving events provides the basis for a logical execution whose dependencies can be reasoned about independently of exact wall-clock timing. Equal physical phases are therefore not required for logical synchrony. Frequency agreement alone, however, does not establish the required event relationships: frame ordering, initialization and continued availability of the corresponding frames also matter. These distinctions follow the logical-synchrony and multiclock-network framework developed by Lall et al. [3].

The control problem supports this realization by regulating physical progress while retaining the prescribed frame relationships. Each node observes its own receiving buffers and adjusts its oscillator. Buffers consequently provide both the local feedback signal and finite storage for a temporary imbalance between delivery and consumption.

2.2 Continuous Clocks and Receiving-Buffer Dynamics

We consider a network of n nodes. Node i has continuous oscillator phase $\theta_i(t)$, measured in local cycles, and physical frequency

$$\dot{\theta}_i(t) = \omega_i(t) = \omega_i^u + u_i(t), \tag{1}$$

where ω_i^u is its constant intrinsic frequency and $u_i(t)$ is the applied frequency correction. The correction can be signed; admissible physical operation requires $\omega_i(t) > 0$. In the current Callisto model, corrections are held between local actuation events. Phase remains continuous when a correction changes, although its slope can change. Local sampling and integer frame events are embedded in this continuous-time plant.

For a receiving edge $e = (j \to i)$, let $\beta_e(t)$ be buffer occupancy in frames and C_e its capacity. During normal operation without progress blocking,

$$\begin{aligned} \beta_e(t) &= \beta_e(0) + A_{ji}^{\mathrm{arr}}(t) - N_i(t), \\ N_i(t) &= \lfloor \theta_i(t) \rfloor - \lfloor \theta_i(0) \rfloor. \end{aligned} \tag{2}$$

Here $A_{ji}^{\mathrm{arr}}(t)$ counts arrivals during $(0, t]$, including frames already in flight at initialization, and $N_i(t)$ counts local consumption. This conservation relation explains why both oscillator mismatch and delivery changes affect occupancy. Section 3 specifies the local schedules, initialization and ordered arrival model used in our realization.

Fluid models replace integer frame counts by continuous phase increments. With constant physical path delay τ_{ji}, the corresponding occupancy satisfies, between command discontinuities,

$$\dot{\widetilde{\beta}}_{ji}(t) = \omega_j(t - \tau_{ji}) - \omega_i(t), \tag{3}$$

where $\widetilde{\beta}_{ji}$ denotes fluid occupancy. Thus occupancy integrates the imbalance between delayed source progress and local consumption. The fluid approximation facilitates analysis; its assumptions must be distinguished from those of integer-frame execution. This separation is also central to existing PI performance analysis [1] and buffer-occupancy modeling [4].

2.3 Safety, Syntony and Buffer Centering

Safety is a property of the complete trajectory. For the normal operating regime studied here, we seek

$$0 < \beta_e(t) < C_e, \qquad \omega_i(t) > 0, \qquad |u_i(t)| \le u_{\max}, \tag{4}$$

for every edge, node and time, where $u_{\max} > 0$ is the correction limit. The available lower and upper buffer reserves are $\beta_e(t)$ and $C_e - \beta_e(t)$, respectively. A convergent trajectory can still exhaust either reserve before reaching its limit. Avoiding such exhaustion through control is distinct from invoking a protection mechanism that blocks logical progress.

Define the physical frequency spread as

$$\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t). \tag{5}$$

For nominal fluid operation with fixed conditions, we seek $\Delta_\omega(t) \to 0$, together with $\beta_e(t) \to \beta_e^*$ for compatible occupancy targets β_e^*. Frequency agreement is the nominal syntonization objective; it does not imply zero phase differences or tracking an externally prescribed frequency. Without an external reference, the common limiting frequency ω_∞ is determined by the closed loop. Likewise, arbitrary per-edge targets cannot be assumed compatible with a fixed marking, meaning the frame-count offsets determined by initialization, and with the selected initial state.

If a transient frequency-spread tolerance $\Delta_{\text{safe}} > 0$ is specified, then $\Delta_\omega(t) < \Delta_{\text{safe}}$ is an additional engineering constraint. It is not the definition of logical synchrony. Under persistent delivery perturbations, we assess bounded deviations, residual disagreement and buffer reserves rather than assume exact disturbance-free centering.

2.4 Delivery Perturbations and Local Feedback

A latency increase affects the delivery process before a receiver necessarily observes it through its next control measurement. If arrivals pause while consumption continues, the conservation relation shows that the receiving buffer decreases; subsequent release of accumulated frames can instead increase occupancy. The disturbance therefore acts through frame availability and accumulated count imbalance. A controller cannot recreate an undelivered frame, and its response is constrained by measurement and actuation timing. Our experiments use declared order-preserving delivery traces so that controller comparisons share the same external stress definition. The exact arrival construction is given in Section 3.

2.5 Existing Control Foundations and the Present Study

Existing work analyzes occupancy-feedback PI performance through controller gains and graph resistance distances, derives limiting fluid occupancies, and develops reframing methods that separate frequency regulation from occupancy positioning. These are complementary foundations: PI performance [1], steady buffer occupancy [4], and buffer centering [2]. Their conclusions apply under their respective modeling assumptions; in particular, continuous-time gain conditions cannot be transferred unchanged to sampled execution with delay.

This paper builds on those foundations by proposing a local incremental Fuzzy PI law and analyzing how its nonlinear increments reshape the closed-loop response at explicitly calibrated PI scales. Section 3 defines the complete controller and establishes that calibration. Section 4 derives sufficient safety and convergence conditions for a declared synchronous, zero-delay fluid reference, followed by an all-time buffer-excursion comparison for one prescribed initialization. Evaluation in continuous-phase Callisto separately examines delivery perturbations under the actual local update and frame-count conventions.

3 Proposed Fuzzy PI Control for bittide and Nonlinear Analysis

We propose a decentralized, occupancy-based incremental Fuzzy PI controller for bittide. Each node combines receiving-buffer error with its update-to-update change to compute an oscillator-correction increment. This section specifies the complete controller and its continuous-clock interface, derives an implementation-consistent PI calibration, and establishes the nonlinear properties used in Section 4. The physical plant, integer frame counts and sampled controller state are kept distinct throughout the derivation.

3.1 Continuous-clock realization of the proposed controller

The plant is the current continuous-phase Callisto model within the bittide framework [3]. For constant intrinsic oscillator frequency ω_i^u and applied correction $u_i(t)$,

$$\begin{aligned} \dot\theta_i(t) &= \omega_i^u + u_i(t), \\ u_i(t) &= u_{i,k}, \qquad t \in [t_{i,k}^a, t_{i,k+1}^a), \\ \theta_i(t) &= \theta_i(t_{i,k}^a) + (\omega_i^u + u_{i,k})(t - t_{i,k}^a). \end{aligned} \tag{6}$$

The superscript a denotes actuation, and k indexes that node's updates. Normal executions require $\omega_i^u + u_i(t) > 0$, so the phase has a monotone inverse. Physical phase is continuous at an update; its slope can change. Requested correction $q_{i,k}$ is computed from the measured buffers and applied as $u_{i,k} = \text{sat}_{u_{\max}}(q_{i,k})$, with $\text{sat}_b(z) = \min\{b, \max\{-b, z\}\}$ and $u_{\max} > 0$. The proposed controller has jump updates inside a continuous physical plant; it is not implemented as an ordinary differential equation for its accumulator.

Let $p > 0$ be the polling period and $d_{\text{ctrl}} \geq 0$ the execution delay, both measured in local-clock phase units. Measurement time $t_{i,k}^m$ and actuation time $t_{i,k}^a$ satisfy

$$\begin{aligned} \theta_i(t_{i,k}^m) &= \theta_i(0) + k\,p, \\ \theta_i(t_{i,k}^a) &= \theta_i(0) + k\,p + d_{\text{ctrl}}, \\ \int_{t_{i,k}^m}^{t_{i,k}^a} (\omega_i^u + u_i(s))\,ds &= d_{\text{ctrl}}. \end{aligned} \tag{7}$$

The product $k\,p$ in these scheduling equations means the integer index k multiplied by the poll period p; it is not the proportional gain k_p. With zero initial correction, $t_{i,0}^m = 0$ and $t_{i,0}^a = d_{\text{ctrl}}/\omega_i^u$. Subsequent actuation intervals equal $p/(\omega_i^u + u_{i,k})$. The delay is not automatically a constant physical time, and its age must not be computed using the new command that is applied only at the end of that age interval.

In normal operation, each integer source-phase crossing sends a frame. With the progress guards open, frame conservation gives

$$\beta_{j\to i}(t) = \beta_{j\to i}(0) + A_{ji}^{\text{arr}}(t) - N_i(t), \tag{8}$$

where $A_{ji}^{\text{arr}}(t)$ counts arrivals during $(0, t]$, including initial in-flight frames, and $N_i(t) = \lfloor \theta_i(t) \rfloor - \lfloor \theta_i(0) \rfloor$ counts local consumption.

For a constant physical delay $\tau_{ji} \geq 0$,

$$A_{ji}^{\mathrm{arr}}(t) = \lfloor \theta_j(t - \tau_{ji}) \rfloor - \lfloor \theta_j(-\tau_{ji}) \rfloor. \tag{9}$$

Thus arrival counts incorporate the declared negative-time history.

For constant physical path delay τ_{ji}, receiving occupancy is

$$\beta_{j\to i}(t) = \lambda_{ji}^{\mathrm{UGN}} + \lfloor \theta_j(t - \tau_{ji}) \rfloor - \lfloor \theta_i(t) \rfloor. \tag{10}$$

Here $\lambda_{ji}^{\mathrm{UGN}}$ is the fixed integer offset in the absolute phase-count convention. It is determined by occupancy and prehistory:

$$\lambda_{ji}^{\mathrm{UGN}} = \beta_{j\to i}(0) - \lfloor \theta_j(-\tau_{ji}) \rfloor + \lfloor \theta_i(0) \rfloor. \tag{11}$$

The relative-count logical marking is $\lambda_{ji}^{\mathrm{LSN}} = \lambda_{ji}^{\mathrm{UGN}} + \lfloor \theta_j(0) \rfloor - \lfloor \theta_i(0) \rfloor$; the two coincide when the initial phase counts are equal.

For changing delivery times, let $s_{j,m} = \theta_j^{-1}(m)$ be the send time of source tick m, and $a_{ji}(s)$ its causal ordered arrival map. For the desired send-time delay profile $\tau_{ji}(s)$, the declared running-maximum envelope is

$$a_{ji}(s) = \sup_{r \leq s} \{ r + \tau_{ji}(r) \},$$

including the constant-delay negative-time history. Define the delivered tick frontier

$$C_{ji}^{\mathrm{arr}}(t) = \max\{ m \in \mathbb{Z} : a_{ji}(s_{j,m}) \leq t \}, \quad \beta_{j\to i}(t) = \lambda_{ji}^{\mathrm{UGN}} + C_{ji}^{\mathrm{arr}}(t) - \lfloor \theta_i(t) \rfloor. \tag{12}$$

The negative-time history specifies initial in-flight frames. The declared running-maximum arrival envelope preserves order and can release a batch after a desired-delay decrease; it imposes no finite serialization rate. At an upward-delay discontinuity, a supremum of continuous source times need not itself have arrived. Counting delivered tick indices avoids silently including such a boundary frame.

If a protection guard blocks logical progress, the runtime can retain a separate continuous data phase ψ_i with $\dot{\psi}_i = g_i(t)\dot{\theta}_i$, where $g_i \in \{0, 1\}$ is the progress gate. Buffer counts then use ψ_i, whereas scheduling still uses θ_i. In the ungated normal regime, with matching initial phases, data phase and physical phase agree. Guard operation does not stop the oscillator.

Only for the reference proof do we replace integer occupancy by a fluid counterpart. At constant delay, define $\widetilde{\beta}_{ji}(t) = \widetilde{\lambda}_{ji} + \theta_j(t - \tau_{ji}) - \theta_i(t)$, where $\widetilde{\lambda}_{ji}$ is an additive offset chosen for the specified fluid initialization. Between command discontinuities,

$$\dot{\widetilde{\beta}}_{ji}(t) = \omega_j^u + u_j(t - \tau_{ji}) - \omega_i^u - u_i(t). \tag{13}$$

This derivative belongs to the fluid counterpart, not the actual integer occupancy. Replacing floor counts by phases does not automatically preserve an arbitrary initialization; the compatible zero-delay marking used in Section 4 is specified there.

3.2 Notation and the measured error

Node i has incoming-neighbour set $\mathcal{N}_i^-$ and $d_i = |\mathcal{N}_i^-| > 0$ incoming channels. All incoming sums below range over $j \in \mathcal{N}_i^-$. This section assumes fixed active incoming sets: the native link-status mask is one on all configured inputs. The Fuzzy percentage wrapper averages its configured inputs, so changing link masks would require an explicit corresponding measurement policy. At the kth local measurement, $\beta_{ji,k}$ is the measured occupancy of channel $j \to i$, and β_{ji}^* is its fixed target. Use

$$E_{i,k} = \sum_{j \in \mathcal{N}_i^-} (\beta_{ji,k} - \beta_{ji}^*). \tag{14}$$

The sign is actual minus target. If incoming buffers are too full, a positive frequency correction increases local consumption. This explains the positive PI gains with this error convention. The physical frequency is $\omega_i = \omega_i^u + u_i$, where ω_i^u is the constant intrinsic frequency; u_i is a correction, not the entire frequency.

Symbol	Meaning
$E_{i,k}$	Sum error in frames
$e_{i,k}$	Target-minus-measured error in percentage points
$I_{i,k}$	Native PI sum of sampled frame errors
$A_{i,k}$	Fuzzy accumulator: total requested correction
$q_{i,k}, u_{i,k}$	Requested and applied frequency corrections
h	Fixed native PI integration scale
$X_{i,k}, Y_{i,k}$	Scaled Fuzzy error and sample-difference inputs
$F_0, \mathcal{R}_F$	Linear counterpart and residual $F - F_0$

The event index k is local to a node. The normalization factor γ_i converts a frame-sum error into percentage points. The maximum correction $u_{\max} > 0$ is the actuator limit; q and u remain distinct if clipping acts. Two nodes need not measure at the same physical times. The subscript ji abbreviates edge $j \to i$. Suppressing i below is a notational convenience, not a synchrony assumption; in that notation $\gamma = \gamma_i$.

3.2.1 Deriving the percentage-to-token relation. For common incoming capacity C, define

$$\bar{p}_{i,k} = \frac{1}{d_i} \sum_j \frac{100\beta_{ji,k}}{C}, \qquad r_i = \frac{1}{d_i} \sum_j \frac{100\beta_{ji}^*}{C}.$$

Then subtraction and factoring give

$$e_{i,k} = r_i - \bar{p}_{i,k} \tag{15}$$

$$= -\frac{100}{d_i C} \sum_j (\beta_{ji,k} - \beta_{ji}^*) = -\gamma_i E_{i,k}, \qquad \gamma_i = \frac{100}{d_i C}. \tag{16}$$

For midpoint targets this is the original absolute-percentage input with $r_i = 50$. For other targets the runtime can supply a centered percentage $r_i + d_i^{-1} \sum_j 100(\beta_{ji,k} - \beta_{ji}^*)/C$; subtracting it from r_i yields the same $e_{i,k}$. A mismatched setpoint instead adds a constant bias and invalidates the scalar relation used for calibration.

With unequal capacities and a consistently matched target reference, the percentage average gives

$$e_{i,k} = -\frac{100}{d_i} \sum_j \frac{\beta_{ji,k} - \beta_{ji}^*}{C_{ji}}.$$

This is a weighted error and generally cannot be written as one normalization constant times the unweighted native sum. Matching such a controller requires a consistent measurement weighting or a different baseline.

3.3 Native PI: update order, units and the first call

The occupancy-feedback PI baseline is grounded in the bittide analysis of [1]. The executed native update is an integrate-then-output recursion:

$$I_k = I_{k-1} + E_k, \qquad q_k^{\mathrm{PI}} = k_p E_k + \kappa_I I_k, \qquad \kappa_I = k_i h. \tag{17}$$

The sum I_k contains no physical time multiplier; h is incorporated in κ_I. In the present runtime,

$$h = \frac{p}{\omega_{\mathrm{base}}}. \tag{18}$$

Here p is the local polling period and ω_{base} is the fixed implementation reference frequency. The scale h stays fixed during the configured run. Local polling intervals measured in physical time can vary as the oscillator changes; replacing h by those intervals would implement a different PI law.

Subtract the previous requested output:

$$q_k^{\mathrm{PI}} - q_{k-1}^{\mathrm{PI}} = k_p(E_k - E_{k-1}) + \kappa_I(I_k - I_{k-1}) \tag{19}$$

$$= k_p(E_k - E_{k-1}) + \kappa_I E_k. \tag{20}$$

This is the correct increment to reproduce. It contains the present E_k in the integral term because integration precedes output.

At zero initial sum, the first call returns

$$q_0^{\mathrm{PI}} = (k_p + \kappa_I)E_0. \tag{21}$$

No physical prehistory measurement is needed to express this algebra: one may use the fictitious previous error $E_{-1} = 0$ and previous requested output $q_{-1} = 0$. That bookkeeping choice is separate from deriving a plant identity such as $x_k - x_{k-1} = Hw_{k-1}$, which requires a genuine elapsed plant interval.

3.4 The proposed Fuzzy PI controller

The proposed local incremental control law combines the present error and its sample difference as follows:

$$X_k = K_e e_k, \qquad Y_k = K_{\Delta e}(e_k - e_{k-1}), \tag{22}$$

$$A_k = A_{k-1} + K_u F(X_k, Y_k), \qquad q_k^{\mathrm{F}} = A_k. \tag{23}$$

The gain signs used here are $K_e, K_{\Delta e} > 0$ and $K_u < 0$. Thus high occupancy produces negative e, and a negative surface output produces a positive correction increment. The quantity $e_k - e_{k-1}$ is not divided by elapsed time; calling it a time derivative would obscure its sampling dependence.

3.4.1 Memberships and rule weights. The labels NL, NS, ZE, PS and PL mean negative large, negative small, zero, positive small and positive large, respectively. Each membership $\varphi_a(z) \in [0, 1]$ is an activation degree; a singleton is a numerical rule consequent. Let $[z]_+ = \max\{z, 0\}$ and $\mathrm{clip}_{[0,1]}\, z$ denote projection onto $[0, 1]$. The exact memberships can be written

$$\varphi_{\mathrm{NL}}(z) = \mathrm{clip}_{[0,1]}\,\frac{-z-5}{5}, \qquad \varphi_{\mathrm{NS}}(z) = [1 - |z+5|/5]_+,$$
$$\varphi_{\mathrm{ZE}}(z) = [1 - |z|/5]_+, \qquad \varphi_{\mathrm{PS}}(z) = [1 - |z-5|/5]_+,$$
$$\varphi_{\mathrm{PL}}(z) = \mathrm{clip}_{[0,1]}\,\frac{z-5}{5}. \tag{24}$$

Table 1 gives all 25 consequent rules. Order the labels as NL, NS, ZE, PS and PL, with label indices $a, b \in \{1, \ldots, 5\}$. With singleton values

Table 1: Rule consequents of the proposed bittide Fuzzy PI controller.

$X \backslash Y$	NL	NS	ZE	PS	PL
NL	NL	NL	NL	NS	ZE
NS	NL	NL	NS	ZE	PS
ZE	NL	NS	ZE	PS	PL
PS	NS	ZE	PS	PL	PL
PL	ZE	PS	PL	PL	PL

$(s_1, \ldots, s_5) = (-5, -2.5, 0, 2.5, 5)$ and $\ell(a, b) = \min\{5, \max\{1, a + b - 3\}\}$, compute

$$w_{ab} = \min\{\varphi_a(X), \varphi_b(Y)\}, \qquad F(X, Y) = \frac{\sum_{a,b} w_{ab} s_{\ell(a,b)}}{\sum_{a,b} w_{ab}}. \tag{25}$$

The membership partition gives at least one positive input membership on each axis, hence a positive denominator at finite inputs. Individual rules are summed even when they share a consequent. Replacing this accumulation by maximum aggregation changes the map and its derivation.

The weighted-average property gives $|F| \le 5$ and therefore $|A_k - A_{k-1}| \le 5|K_u|$. This is a per-update bound, not a bound on A_k over an arbitrarily long execution. The raw Fuzzy module itself has no final correction clipping. Initialize $A_{i,-1} = 0$ and $e_{i,-1} = 0$. At each measurement, form $e_{i,k}$ and $(X_{i,k}, Y_{i,k})$, evaluate the rule weights and normalized surface, increment $A_{i,k}$, and store $e_{i,k}$ as the previous error. The requested correction is then limited and applied at $t_{i,k}^a$, and held until the next actuation. In the normal-run policy considered here, external clipping retains the raw accumulator and anti-windup is disabled. The controller does not overwrite $A_{i,k}$ with the applied command. The membership functions, rule consequents and gains stay fixed during a run; the correction changes with the measured input pair. The law requires neither an observer nor an FFP mode.

3.5 Exact PI Calibration, Including Initialization

Define the globally linear counterpart

$$F_0(X, Y) = \frac{X + Y}{2}. \tag{26}$$

Replacing F by this map retains the same controller accumulator. Using $e = -\gamma E$ gives

$$A_k^0 - A_{k-1}^0 = \frac{K_u}{2}\{K_e e_k + K_{\Delta e}(e_k - e_{k-1})\} \tag{27}$$

$$= -\frac{K_u \gamma K_e}{2} E_k - \frac{K_u \gamma K_{\Delta e}}{2}(E_k - E_{k-1}). \tag{28}$$

Matching the two independent coefficients in (20) yields

$$k_p = -\frac{K_u \gamma K_{\Delta e}}{2}, \qquad \kappa_I = -\frac{K_u \gamma K_e}{2}. \tag{29}$$

Solving for two Fuzzy gains with chosen $K_e > 0$ gives

$$K_u = -\frac{2k_i h}{\gamma K_e}, \qquad K_{\Delta e} = \frac{K_e k_p}{k_i h}. \tag{30}$$

The formulas assume $k_i h > 0$. Pure proportional control is a separate limiting case and should not be inserted into this inverse mapping with a zero denominator.

Proposition 3.1 (Event-level linear-counterpart equivalence). *For any measurement sequence, suppose* (29) *holds and initialize*

$$A^0_{-1} = k_p E_{-1} + \kappa_I I_{-1}, \qquad e_{-1} = -\gamma E_{-1}. \tag{31}$$

Then $A^0_k = k_p E_k + \kappa_I I_k$ *for every* $k \geq 0$.

Proof. Assume the relation holds before update k. Substitution gives

$$\begin{aligned} A^0_k &= k_p E_{k-1} + \kappa_I I_{k-1} + k_p(E_k - E_{k-1}) + \kappa_I E_k \\ &= k_p E_k + \kappa_I (I_{k-1} + E_k) = k_p E_k + \kappa_I I_k. \end{aligned}$$

The stated initialization supplies the induction base, including $k = 0$. □

3.5.1 Why zero constructors work for a nonzero first error. Take $I_{-1} = A^0_{-1} = e_{-1} = 0$, corresponding to $E_{-1} = 0$. The first Fuzzy-reference input is

$$(X_0, Y_0) = (-\gamma K_e E_0, -\gamma K_{\Delta e} E_0),$$

so $A^0_0 = (k_p + \kappa_I)E_0$, exactly as in (21). If only Fuzzy is warmed up to $e_{-1} = e_0$ while $A^0_{-1} = 0$, its difference input vanishes and the output is merely $\kappa_I E_0$. The missing $k_p E_0$ term can create a large apparent controller difference unrelated to fuzzy inference.

3.5.2 A numerical calibration from the present configuration. For $d = 2$, $C = 2048$, $p = 1000$, $\omega_{\text{base}} = 29999/30000$, $k_p = 2 \times 10^{-6}$, $k_i = 5 \times 10^{-11}$ and $K_e = 0.1$,

$$\begin{aligned} \gamma &= \frac{25}{1024}, & h &= \frac{30000000}{29999}, \\ \kappa_I &= \frac{3}{59998000}, & K_u &= -\frac{768}{18749375}, \\ K_{\Delta e} &= \frac{29999}{7500} \simeq 3.9998666667. \end{aligned} \tag{32}$$

For a one-frame first error, the matched PI and F_0 corrections are both $2.0500016667 \times 10^{-6}$. Warming up only the Fuzzy previous error instead gives approximately $5.0001666722 \times 10^{-8}$. The example illustrates an initialization mismatch; it is not a new plant simulation.

3.5.3 Execution-layer conditions. Identical raw outputs give identical applied outputs under a common limiter when raw states are retained and anti-windup is disabled. Tracking or reset requires compatible state changes. For example, after processing E_k, setting

$$A^0_k = u_{\text{held}}, \qquad I_k = \frac{u_{\text{held}} - k_p E_k}{\kappa_I}, \qquad e_k = -\gamma E_k$$

restores the state relation for the next ordinary update. A branch designed to produce a held output on an immediate extra update requires its own integrate-first adjustment; the above is not an instruction to add an extra call. Limiting the internal reference F_0 to $[-5, 5]$ also changes the controller, even if the external correction limiter is shared.

Under a deterministic causal plant, the same initial history, measured inputs, actuation order, external trace and compatible resets imply closed-loop equality by induction over events. No globally synchronous physical grid is required for this algebraic equivalence. By contrast, matching only labels such as "same gains" or "same random seed" does not establish these conditions.

3.6 Deriving the Exact Central Nonlinear Surface

For $0 < r \leq 5$, define

$$\begin{aligned} \Omega_r &= \{(X, Y) : |X| + |Y| \leq r\}, \\ m &= \min(|X|, |Y|), \qquad s = |X| + |Y|. \end{aligned} \tag{33}$$

The same-sign interaction is

$$J(X, Y) = \begin{cases} \operatorname{sgn}(X)m, & XY > 0, \\ 0, & XY \leq 0. \end{cases} \tag{34}$$

For points in Ω_5, put $a = |X|/5$, $b = |Y|/5$. Because $a + b \leq 1$, each input has only its central and sign-adjacent membership active. Their four joint min weights are

$$\begin{aligned} w_{00} &= \min(1 - a, 1 - b) = 1 - \max(a, b), \\ w_{s0} &= \min(a, 1 - b) = a, \\ w_{0s} &= \min(1 - a, b) = b, \\ w_{ss} &= \min(a, b). \end{aligned} \tag{35}$$

The subscripts s denote the input's active sign label, not the scalar $s = |X| + |Y|$. Zero-input boundaries follow by continuity.

The denominator is therefore

$$D = 1 - \max(a, b) + a + b + \min(a, b) = 1 + 2\min(a, b) = 1 + 2m/5. \tag{36}$$

The central-central consequent is zero. The sign-central and central-sign rules contribute $(5/2)\operatorname{sgn}(X)a = X/2$ and $(5/2)\operatorname{sgn}(Y)b = Y/2$. For equal signs, the joint consequent is $5\operatorname{sgn}(X)$ and contributes $\operatorname{sgn}(X)m$; for opposing signs it is zero. The four quadrants therefore give

$$F(X, Y) = \begin{cases} \dfrac{F_0 + m}{1 + 2m/5}, & X > 0,\ Y > 0, \\ \dfrac{F_0 - m}{1 + 2m/5}, & X < 0,\ Y < 0, \\ \dfrac{F_0}{1 + 2m/5}, & X > 0,\ Y < 0, \\ \dfrac{F_0}{1 + 2m/5}, & X < 0,\ Y > 0. \end{cases} \tag{37}$$

On either axis $m = 0$, giving $F = F_0$. These cases establish the following identity, including boundaries by continuity.

Proposition 3.2 (Exact central surface). *For every* $(X, Y) \in \Omega_5$,

$$F(X, Y) = \frac{(X + Y)/2 + J(X, Y)}{1 + 2m/5}, \qquad J(X, Y) = \begin{cases} \operatorname{sgn}(X)m, & XY > 0, \\ 0, & XY \leq 0. \end{cases} \tag{38}$$

This is an exact rule-level identity on Ω_5, not a fitted approximation. Outside this diamond, the cross weights in (35) need not equal a and b, so the formula must not be extended without a new case analysis.

3.7 Sign-conditioned response, sector and nonlinear geometry

3.7.1 Same-sign enhancement. When $XY > 0$, F_0 has magnitude $s/2$ and the same sign as X. Equation (38) gives

$$\frac{F}{F_0} = \frac{1 + 2m/s}{1 + 2m/5}. \tag{39}$$

Since $s \le 5$, its numerator is at least the denominator. Since $m \le s/2$, the numerator is at most 2. Thus $1 \le F/F_0 \le 2$. Define the normalized nonlinear residual by $\mathcal{R}_F(X, Y) = F(X, Y) - F_0(X, Y)$. Subtracting F_0 gives

$$\mathcal{R}_F = \frac{\operatorname{sgn}(X) m(1 - s/5)}{1 + 2m/5}, \qquad XY > 0. \tag{40}$$

For instance, $F(0.1, 0.1) = 0.2/1.04 \simeq 0.19230769$, compared with $F_0 = 0.1$. Both are surface values; the actual increment multiplies by negative K_u.

3.7.2 Opposing-sign attenuation and zero-output cases. When $XY < 0$, the interaction vanishes, so

$$F = \frac{F_0}{1 + 2m/5}, \qquad \mathcal{R}_F = -\frac{2m/5}{1 + 2m/5} F_0. \tag{41}$$

For $(X, Y) = (0.1, -0.05)$, $F_0 = 0.025$ and $F = 0.025/1.02 \simeq 0.024509804$. On either axis $m = 0$ and $F = F_0$. On the anti-diagonal $Y = -X$, both outputs are exactly zero; a ratio should not be evaluated there.

Corollary 3.3 (Regional pointwise sector). *For $(X, Y) \in \Omega_r$, $0 < r \le 5$, one can write*

$$F = \alpha(X, Y) F_0, \qquad \underline{\alpha}_r = \frac{1}{1 + r/5} \le \alpha(X, Y) \le 2. \tag{42}$$

Proof. Same-sign ratios lie in $[1, 2]$. For opposing signs with $F_0 \neq 0$, $m \le r/2$ gives $1/(1 + r/5) \le F/F_0 \le 1$. The axes agree with F_0. If $F_0 = 0$, both outputs vanish and any admissible α preserves the equality. □

The equivalent division-free inequality is

$$[F - \underline{\alpha}_r F_0][2F_0 - F] \ge 0. \tag{43}$$

In particular, Ω_1 gives $\alpha \in [5/6, 2]$, while Ω_5 gives $[1/2, 2]$. These bounds describe pointwise values of the specific nonlinear map. They are not a gain-scheduled stability proof; the closed loop must retain the input region and accommodate different node-dependent, time-varying α values.

3.7.3 Why an axis match is not a Jacobian linearization. The same-sign term J is positively homogeneous. For $t > 0$ and a fixed direction $v = (v_1, v_2)$, the central formula yields

$$F'(0; v) = \lim_{t \downarrow 0} \frac{F(tv_1, tv_2)}{t} = \frac{v_1 + v_2}{2} + J(v_1, v_2). \tag{44}$$

Consequently $F'(0; (1, 0)) = F'(0; (0, 1)) = 1/2$, but $F'(0; (1, 1)) = 2$. A linear derivative with the two axis values would instead give 1 on $(1, 1)$, a contradiction. The map is nonlinear and not Fréchet differentiable at the origin, despite exact agreement with F_0 on the coordinate axes. "Axis-calibrated linear counterpart" is therefore the appropriate term.

3.7.4 Where the central interaction comes from. As an analytic counterfactual, replace min rule weights by $w_{ab}^{\text{prod}} = \varphi_a(X)\varphi_b(Y)$, retaining the membership functions and singletons. On $|X|, |Y| \le 5$, the active consequents add their two single-input values and the denominator is $(\sum_a \varphi_a(X))(\sum_b \varphi_b(Y)) = 1$. Thus

$$F_{\text{prod}}(X, Y) = X/2 + Y/2 = F_0(X, Y). \tag{45}$$

The central nonlinear interaction is tied to normalized min activation, not merely to linguistic labels. The identity isolates the role of min activation; it does not assert linearity beyond the central square or an experimental comparison of product and min inference.

3.8 Nonlinear memory and what it does not identify

For identical scaled input pairs (X_k, Y_k), including consistent previous-error initialization, let A_k^{F} use F and A_k^0 use F_0, with equal initial accumulators. Subtracting their updates gives

$$A_k^{\text{F}} - A_k^0 = (A_{k-1}^{\text{F}} - A_{k-1}^0) + K_u \mathcal{R}_F(X_k, Y_k).$$

Telescoping yields

$$A_k^{\text{F}} - A_k^0 = K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{46}$$

With zero native/Fuzzy constructor states and the calibrated parameters, the same-input bookkeeping can also be written

$$A_k^{\text{F}} = k_p E_k + \kappa_I \sum_{\ell=0}^{k} E_\ell + K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{47}$$

The first two terms form a PI-like shadow evaluated on the Fuzzy trajectory's own inputs. The last term records accumulated nonlinear correction. If later inputs lie on the axes, the new residual is zero but the accumulated difference can remain nonzero.

This identity requires no accumulator projection/reset. Pure external clipping with retained raw states leaves the raw identity intact, but the applied difference must be evaluated through the limiter. Tracking/reset requires explicit additional state-change terms. Moreover, independently evolving PI and Fuzzy closed loops generate different measurements and possibly different physical event schedules. A same-input shadow identity therefore cannot be relabelled as their observed closed-loop difference.

3.9 Interpretation and the bridge to the stability section

The derivation provides implementation-consistent PI calibration, the exact central nonlinear map, and a regional sector suitable for a closed-loop certificate. Section 4 uses these properties after stating its plant, sampling, latency and initialization assumptions and proving that the sector's input region remains invariant. The present algebra does not change the continuous-clock Callisto plant into a discrete-time plant; it describes the controller updates embedded in that execution.

Same-sign scaled inputs indicate alignment of the two input channels. They do not always mean increasing absolute error: a transition from $e_{k-1} = -10$ to $e_k = 1$ has $e_k(e_k - e_{k-1}) > 0$ even though the absolute error decreased. Likewise, opposing-sign attenuation concerns a single increment relative to F_0, not an established

network jitter attenuation property. Frequency disagreement, buffer excursion, common frequency and control activity must be evaluated separately.

4 Reference-Model Convergence and Buffer Margins

We connect the nonlinear controller of Section 3 to four separate properties: invariance of its analyzed input region, convergence of frequency and buffer disagreements, strict buffer-capacity safety, and positive physical oscillator frequency. We then establish a strict headroom comparison with its calibrated PI baseline for one specified initialization. The proof concerns an explicitly defined continuous fluid reference. Actual Callisto retains asynchronous local sampling, delivery delays and integer frame events, which are not silently removed from its experimental model.

4.1 Reference assumptions and continuous-time model

4.1.1 What the reference retains and idealizes. The reference retains the controller of Section 3, continuous oscillator phases, local occupancy feedback, the native integrate-before-output PI convention and the proposed incremental normalized-min Fuzzy surface. It assumes zero path delay, zero control-execution delay, fluid rather than integer occupancy and a common physical sampling grid $t_k = kH$, $H > 0$. A zero-order hold (ZOH) means that the applied frequency correction is constant on each interval $[t_k, t_{k+1})$. These are explicit analysis assumptions; local-clock polling in actual Callisto is not asserted to have this common grid. The equations sampled at kH below are exact endpoint relations for that continuous ZOH flow. They are neither numerical Euler integration nor a discrete firing-count replacement of the plant.

Let $\mathcal{N}_i^-$ be the fixed incoming-neighbour set and $d_i = |\mathcal{N}_i^-|$. The graph is connected, undirected and unit-weight, with a receiving channel in both directions. Its symmetric Laplacian L has diagonal d_i and off-diagonal -1 at neighbouring nodes. Each undirected connection contributes once to this Laplacian; the pair of directed receiving channels does not double its weights.

The physical phase is θ_i, the intrinsic frequency is ω_i^u, the applied correction is u_i, and the requested correction is q_i. The target occupancy of receiving edge $j \to i$ is $\beta^*_{j\to i}$. The fixed coefficient scale in the native PI law is $h = p/\omega_{\text{base}}$, where p is a local polling period and ω_{base} is the implementation reference frequency; H is the physical sampling period imposed only in this reference. The integral coefficient is $\kappa_I = k_i h$. These two time scales are kept distinct. All receiving buffers have common capacity $C > 0$, and the intrinsic frequencies ω_i^u are constant. The all-ones column vector is $\mathbf{1}$.

The transpose is denoted by $\top$. A symmetric matrix $A \succeq 0$ is positive semidefinite; $A \succ 0$ is positive definite. The matrix order $A \preceq B$ means $B - A \succeq 0$. The symbol $\otimes$ denotes the Kronecker product, and $\|z\|_\infty = \max_i |z_i|$.

4.1.2 Compatible occupancy targets and marking. Throughout the proved centered reference, all initial phases are equal. For zero delay the fluid law in the additive phase-offset convention is

$$\beta_{j\to i}(t) = \theta_j(t) - \theta_i(t) + \lambda_{j\to i}. \tag{48}$$

Here $\lambda_{j\to i}$ is the fixed additive phase offset. With equal initial phases it also equals the initial marking; this presentation does not identify these conventions for arbitrary unequal initial phases. The law follows the bittide fluid modelling framework [1, 3]. Define $\beta^*_{j\to i} = \theta_j(0) - \theta_i(0) + \lambda_{j\to i}$, so the reference starts at its compatible targets. For a common constant ramp frequency ω_{ref} put

$$x_k = \theta(t_k) - \theta(0) - t_k \omega_{\text{ref}} \mathbf{1}. \tag{49}$$

The common ramp cancels from edge differences. Subtracting the target in (48) gives

$$\beta_{j\to i}(t_k) - \beta^*_{j\to i} = x_{j,k} - x_{i,k}, \qquad E_k = -Lx_k. \tag{50}$$

The vector E collects the Section 3 frame errors, actual minus target. Not every set of per-edge targets is compatible with a fixed marking and phase initialization; that compatibility is an assumption of this model.

For the worked example, equal initial phases and target 1024 imply $\lambda_{j\to i} = 1024$. The actual delayed runs have marking 1044 with 20 initial in-flight frames; simply retaining 1044 while imposing zero delay and the same centered occupancy would change the initialization.

4.2 From the original controller to the sampled recurrence

Define the intrinsic offsets $\delta\omega = \omega^u - \omega_{\text{ref}}\mathbf{1}$ and the physical frequency offsets $w_k = \delta\omega + u_k$. For $k \geq 1$, let $v_k = Hw_{k-1}$. The plant's previous genuine interval gives

$$x_k - x_{k-1} = Hw_{k-1} = v_k. \tag{51}$$

Indeed, on each zero-order-held interval,

$$x(t) = x_k + (t - t_k)w_k, \qquad t \in [t_k, t_{k+1}], \tag{52}$$

where $x(t) = \theta(t) - \theta(0) - t\omega_{\text{ref}}\mathbf{1}$. Integration gives $x_{k+1} = x_k + Hw_k$ exactly. The controller is updated only at the endpoints, but the plant evolves continuously throughout the interval.

The factor H is physical elapsed time; the controller factor h remains the native fixed coefficient scale. Section 3 uses $e_i = -\gamma_i E_i$, so the Fuzzy inputs are

$$X_k = K_e \Gamma L x_k, \qquad Y_k = K_{\Delta e} \Gamma L v_k, \qquad \Gamma = \operatorname{diag}(\gamma_1, \ldots, \gamma_n). \tag{53}$$

Here $K_e, K_{\Delta e}$ are chosen common for readability; node-dependent output scaling can still produce common calibrated k_p, k_i.

On $\Omega_r = \{(X, Y) : |X| + |Y| \leq r\}$, $0 < r \leq 5$, Section 3 gives $F = \alpha F_0$, where $F_0 = (X + Y)/2$ and $\underline{\alpha} = (1 + r/5)^{-1} \leq \alpha \leq \overline{\alpha} = 2$. With $D_{\alpha,k} = \operatorname{diag}(\alpha_{1,k}, \ldots, \alpha_{n,k})$, the calibrated increment becomes

$$u_k - u_{k-1} = -D_{\alpha,k}(\kappa_I L x_k + k_p L v_k). \tag{54}$$

Multiplication by H and the next plant interval yield

$$\begin{aligned} v_{k+1} &= v_k - D_{\alpha,k}(bLx_k + aLv_k), \\ x_{k+1} &= x_k + v_{k+1}, \qquad a = Hk_p, \quad b = H\kappa_I = Hk_i h. \end{aligned} \tag{55}$$

The extra H in b is essential: v is a scaled frequency. The surface factor $\alpha_{i,k}$ is node-dependent and state-dependent; freezing it and checking one constant-gain PI characteristic is insufficient.

4.2.1 The initial constructor call. The controller's stored previous error is not a previous plant interval. For the centered initialization $x_0 = 0$ with zero PI sample sum, zero Fuzzy accumulator and zero previous percentage error, the first original controller call gives $u_0 = 0$. Advancing $[0, H]$ then gives

$$x_1 = H\delta\omega, \qquad v_1 = H\delta\omega. \tag{56}$$

Only from $k = 1$ does (51) justify the recurrence. Nonzero initial occupancy error is outside this centered formulation: one must also introduce target-compatible phase coordinates or retain the affine error bias, then explicitly evaluate the original constructor and recheck the initial state and certificate. Evaluating the first call alone does not restore $E = -Lx$. The centered bootstrap is consistent with, but separate from, the event-level first-call PI/F_0 equivalence proved in Section 3.

4.3 Disagreement projection and the symmetric sector matrix

Since $L\mathbf{1} = 0$, a common phase shift does not change occupancies. Choose U with orthonormal columns spanning the subspace orthogonal to $\mathbf{1}$, so $U^\top U = I_{n-1}$ and $U^\top \mathbf{1} = 0$. Choose its columns as graph eigenvectors, giving

$$L = U\Lambda U^\top, \qquad \Lambda = \operatorname{diag}(\mu_2, \dots, \mu_n), \quad 0 < \mu_2 \le \mu_n. \tag{57}$$

I_{n-1} denotes the identity matrix of dimension $n-1$. Define

$$\xi_k = \Lambda^{1/2} U^\top x_k, \qquad \eta_k = \Lambda^{1/2} U^\top v_k, \qquad y_k = (\xi_k^\top, \eta_k^\top)^\top. \tag{58}$$

The symbol ξ is projected phase and η already contains the frequency scale H. The coordinates retain the physical-time scale H; in particular, η is a scaled frequency disagreement rather than an integral-controller state.

For $M_k = \Lambda^{1/2} U^\top D_{\alpha,k} U \Lambda^{1/2}$, substitution in (55) gives

$$\begin{aligned} \eta_{k+1} &= -bM_k\xi_k + (I_{n-1} - aM_k)\eta_k, \\ \xi_{k+1} &= (I_{n-1} - bM_k)\xi_k + (I_{n-1} - aM_k)\eta_k. \end{aligned} \tag{59}$$

In particular, the diagonal D_α need not commute with L; the projection produces the symmetric M_k instead of assuming shared modes. Its quadratic form satisfies

$$\underline{\alpha}\Lambda \preceq M_k \preceq \overline{\alpha}\Lambda, \qquad \operatorname{eig}(M_k) \subseteq [\nu_-, \nu_+] = [\underline{\alpha}\mu_2, \overline{\alpha}\mu_n]. \tag{60}$$

The scalar ν used below is a possible eigenvalue of M, not a node frequency or the number of communication edges.

4.4 A common Lyapunov certificate from two endpoints

Define For matrices $M \in \mathbb{R}^{(n-1)\times(n-1)}$ and scalars ν, define

$$\mathcal{B}(M) = \begin{bmatrix} I - bM & I - aM \\ -bM & I - aM \end{bmatrix}. \tag{61}$$

The identity blocks have dimension $n-1$. Equations above give $y_{k+1} = \mathcal{B}(M_k) y_k$. A scalar block associated with an eigenvalue of M_k is

$$B(\nu) = \begin{bmatrix} 1 - b\nu & 1 - a\nu \\ -b\nu & 1 - a\nu \end{bmatrix}. \tag{62}$$

For a symmetric positive-definite matrix $\bar{P} \in \mathbb{R}^{2\times 2}$, put $P = \bar{P} \otimes I_{n-1}$. The scalar $V_k = y_k^\top P y_k$ is a quadratic measure of phase and frequency disagreement, not of absolute phase.

Lemma 4.1 (Endpoint certificate). *Suppose* $0 < \rho < 1$ *and*

$$\begin{aligned} B(\nu_-)^\top \bar{P} B(\nu_-) &\preceq \rho\bar{P}, \\ B(\nu_+)^\top \bar{P} B(\nu_+) &\preceq \rho\bar{P}. \end{aligned} \tag{63}$$

Then every update whose inputs satisfy the regional sector obeys

$$V_{k+1} \le \rho V_k. \tag{64}$$

The conclusion permits node-dependent and time-varying sector factors.

Proof. First consider one possible scalar eigenvalue. For $\nu = (1-t)\nu_- + t\nu_+$, $0 \le t \le 1$, affine dependence gives $B(\nu) = (1-t)B_- + tB_+$, where $B_\pm = B(\nu_\pm)$. Expanding the two quadratic expressions yields

$$\begin{aligned} &(1-t)B_-^\top \bar{P} B_- + tB_+^\top \bar{P} B_+ - B(\nu)^\top \bar{P} B(\nu) \\ &\qquad = t(1-t)(B_+ - B_-)^\top \bar{P}(B_+ - B_-) \succeq 0. \end{aligned} \tag{65}$$

Together with the endpoint assumptions, this proves $B(\nu)^\top \bar{P} B(\nu) \preceq \rho\bar{P}$ throughout the continuous interval $[\nu_-, \nu_+]$. No eigenvalue grid is required.

Next, each M_k is real symmetric, so it admits an orthogonal factorization $M_k = Q_k \operatorname{diag}(\nu_{1,k}, \dots, \nu_{n-1,k}) Q_k^\top$. Rotating both ξ_k and η_k by $Q_k^\top$ reduces the dynamics to the scalar blocks $B(\nu_{\ell,k})$. The corresponding block rotation leaves $P = \bar{P} \otimes I_{n-1}$ unchanged. Thus summing the scalar inequalities gives $\mathcal{B}(M_k)^\top P \mathcal{B}(M_k) \preceq \rho P$, which is (64). Because the witness P is the same for every permitted M_k, Q_k may change at every step. No simultaneous diagonalization of the node-dependent matrices is assumed. □

The remaining task is to show that the actual nonlinear inputs stay in the region where the sector was obtained. Stability of a family of hypothetical linear blocks alone would not establish that fact.

4.5 Closing the invariant input region

Let $\mathbf{e}_i$ denote the ith unit vector and define the row $a_i^\top = \mathbf{e}_i^\top U \Lambda^{1/2}$. Its squared Euclidean norm is $\|a_i\|^2 = L_{ii} = d_i$. The Fuzzy inputs become

$$X_{i,k} = \gamma_i K_e a_i^\top \xi_k, \qquad Y_{i,k} = \gamma_i K_{\Delta e} a_i^\top \eta_k. \tag{66}$$

For any vector c and $P \succ 0$, Cauchy–Schwarz in the quadratic metric gives $(c^\top y)^2 \le (y^\top P y)(c^\top P^{-1} c)$. Also $|X| + |Y| = \max_{\sigma=\pm1} |X + \sigma Y|$. Applying that output bound on $V \le R$ yields

$$\begin{aligned} |X_{i,k}| + |Y_{i,k}| &\le \sqrt{R\Psi_i}, \\ \Psi_i &= \gamma_i^2 d_i \max_{\sigma=\pm1} [K_e, \sigma K_{\Delta e}] \bar{P}^{-1} [K_e, \sigma K_{\Delta e}]^\top. \end{aligned} \tag{67}$$

The scalar Ψ_i is the largest squared input gain on the energy ellipsoid. If $V_1 \le R$ and $R\Psi_i \le r^2$ for every node, then: each node's first scaled input pair belongs to Ω_r; the valid sector gives $V_2 \le \rho V_1 \le R$; the same output bound keeps the next inputs in Ω_r; induction repeats. This proves the sector domain and disagreement ellipsoid invariant together, rather than assuming either.

Equivalently, a valid radius must satisfy

$$V_1 \le R \le \min_i \frac{r^2}{\Psi_i}. \tag{68}$$

The initialization, normalization through d_i and C, the Fuzzy input scales and the graph-dependent witness therefore enter the same condition. When $\Psi_i = 0$, that node places no restriction on R;

otherwise the displayed quotient is finite. The inference surface is still nonlinear inside the invariant region. The sector is an enclosure of its instantaneous response, not an assertion that its gain is fixed.

Because $P \succ 0$ and $V_k \le \rho^{k-1} V_1$, the projected phase and frequency decay exponentially. The ellipsoid constrains disagreement; the common physical phase continues to grow and is not bounded by V. Thus convergence means agreement of rates and restoration of compatible occupancy targets, not convergence of the absolute oscillator phases to constants.

4.6 From disagreement to physical buffer and clock bounds

4.6.1 Edge occupancy and intersample time. For edge $e = (j \to i)$, define its signed unit incidence vector $\mathbf{b}_e = \mathbf{e}_j - \mathbf{e}_i$. Since it is orthogonal to $\mathbf{1}$, its output from the projected phase is

$$\mathbf{b}_e^\top x_k = \mathbf{b}_e^\top U \Lambda^{-1/2} \xi_k.$$

Define the graph pseudoinverse $L^\dagger = U\Lambda^{-1}U^\top$ and effective resistance $R_e^{\mathrm{eff}} = \mathbf{b}_e^\top L^\dagger \mathbf{b}_e$. The quadratic output inequality gives

$$|\beta_e(t_k) - \beta_e^*|^2 \le V_k [\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}}. \tag{69}$$

Under ZOH the phase, hence each fluid occupancy, is affine on an interval. For $0 \le s \le 1$ its deviation is a convex combination of endpoint deviations, so its absolute value cannot exceed the larger endpoint absolute value. This covers all physical time, including the initial interval from the zero state to x_1. It would not apply unchanged to arbitrary delayed integer arrival events.

Corollary 4.2 (Buffer safety within the reference model). *Suppose the compatible centered initialization and the invariant-domain conditions above hold with $V_1 \le R$. For each receiving edge define*

$$m_e = \min\{\beta_e^*, C - \beta_e^*\},$$
$$B_e(R) = \sqrt{R[\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}}}. \tag{70}$$

If $0 < \beta_e^* < C$ and $B_e(R) < m_e$ on every receiving edge, then the unconstrained fluid execution satisfies $0 < \beta_e(t) < C$ for all $t \ge 0$. In addition, $\beta_e(t) - \beta_e^* \to 0$ as $t \to \infty$.

Proof. At t_k, the squared deviation is bounded by $V_k[\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}}$ and hence by $B_e(R)^2$. The initial deviation is zero. On $[t_k, t_{k+1}]$, the deviation is a convex combination of its endpoint deviations, so its absolute value is at most $B_e(R)$. The strict headroom inequalities give the lower and upper capacity inequalities. Finally $V_k \le \rho^{k-1} V_1 \to 0$ and the same interpolation argument imply convergence at every physical time, not only at samples. □

Here $B_e(R) < m_e$ is a sufficient condition. If it fails, the certificate is inconclusive; failure of this conservative test does not prove a buffer overflow or underflow. The condition also concerns the unconstrained flow. It does not identify a protection-induced stall with successful control convergence.

4.6.2 Frequency-difference safety and convergence. Define the physical frequency spread by

$$\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t). \tag{71}$$

A small frequency spread means close oscillator rates; it does not require the phases or their integer counts to coincide. For any pair $i \ne j$, not necessarily adjacent, define

$$R_{ij}^{\mathrm{eff}} = (\mathbf{e}_i - \mathbf{e}_j)^\top L^\dagger (\mathbf{e}_i - \mathbf{e}_j), \qquad R_{\max}^{\mathrm{eff}} = \max_{i \ne j} R_{ij}^{\mathrm{eff}}, \tag{72}$$

where $\mathbf{e}_i$ is the ith unit vector. Connectedness makes these quantities finite. During $[t_k, t_{k+1})$, $\omega_i(t) = \omega_{\mathrm{ref}} + w_{i,k}$, while $v_{k+1} = H w_k$. Therefore

$$\begin{aligned} |\omega_i(t) - \omega_j(t)|^2 &= H^{-2} |(\mathbf{e}_i - \mathbf{e}_j)^\top v_{k+1}|^2 \\ &\le H^{-2} V_{k+1} [\bar{P}^{-1}]_{22} R_{ij}^{\mathrm{eff}} \\ &\le H^{-2} R \rho^k [\bar{P}^{-1}]_{22} R_{ij}^{\mathrm{eff}}. \end{aligned} \tag{73}$$

The factor V_{k+1}, rather than V_k, is essential: v_k represents the frequency held during the previous interval. The case $k = 0$ is covered by the explicitly constructed bootstrap state y_1. At switching times the same bound applies to the newly held frequency; the previous interval bounds the left-hand limit.

Corollary 4.3 (A prescribed frequency-spread threshold). *Under the invariant-domain conditions, let $\Delta_{\mathrm{safe}} > 0$ be a specified engineering tolerance on pairwise frequency differences. If*

$$D(R) := \frac{1}{H} \sqrt{R[\bar{P}^{-1}]_{22} R_{\max}^{\mathrm{eff}}} < \Delta_{\mathrm{safe}}, \tag{74}$$

then for every $t \in [t_k, t_{k+1})$, $k \ge 0$,

$$\Delta_\omega(t) \le D(R) \rho^{k/2} < \Delta_{\mathrm{safe}}, \qquad \Delta_\omega(t) \longrightarrow 0. \tag{75}$$

Proof. The spread equals $\max_{i,j} |\omega_i - \omega_j|$. Maximizing (73) and taking square roots proves the bound. Since $0 < \rho < 1$, the bound tends to zero. □

The threshold is a user-specified performance or hardware tolerance; logical synchrony does not by itself prescribe its numerical value. This proof bounds the transient as well as the limiting disagreement. It neither proves a smaller Fuzzy frequency peak than PI nor adds a guarantee for arbitrary delayed or stochastic executions. Frequency spread also cannot certify positive absolute frequency: a collection of equal nonpositive rates would have zero spread. The common frequency must therefore be bounded separately.

4.6.3 The unprojected common frequency. Let $\bar{w}_k = n^{-1} \mathbf{1}^\top w_k$ be the common-frequency offset. Put $\alpha_c = (\overline{\alpha} + \underline{\alpha})/2$ and $s_\alpha = (\overline{\alpha} - \underline{\alpha})/2$. With $g = (\kappa_I, k_p)^\top$, averaging (54) and using $\mathbf{1}^\top U = 0$ gives

$$\bar{w}_k - \bar{w}_{k-1} = -\frac{1}{n} \mathbf{1}^\top (D_{\alpha,k} - \alpha_c I_n) U \Lambda^{1/2} (\kappa_I \xi_k + k_p \eta_k). \tag{76}$$

The second component of g is k_p, not $H k_p$, since η already contains H. Bounding each factor yields

$$|\bar{w}_k - \bar{w}_{k-1}| \le s_\alpha \sqrt{\frac{\mu_n}{n} g^\top \bar{P}^{-1} g} \sqrt{V_k}. \tag{77}$$

The geometric series in $\sqrt{\rho}$ is summable. Hence

$$\begin{aligned} \sup_k |\bar{w}_k| &\le |\bar{w}_0| + \frac{s_\alpha \sqrt{(\mu_n/n) R g^\top \bar{P}^{-1} g}}{1 - \sqrt{\rho}} \\ &\le |\bar{w}_0| + \frac{2 s_\alpha}{1 - \rho} \sqrt{\frac{\mu_n}{n} R g^\top \bar{P}^{-1} g} = W_c. \end{aligned} \tag{78}$$

The last inequality follows from $(1 - \sqrt{\rho})^{-1} = (1 + \sqrt{\rho})/(1 - \rho) \le 2/(1 - \rho)$. Absolute summability also proves convergence of

$\bar{w}_k$. Combined with vanishing disagreement, all node frequencies converge to one value $\omega_\infty = \omega_{\mathrm{ref}} + \lim_k \bar{w}_k$. For Fuzzy PI that limit need not be the initial intrinsic mean.

4.6.4 Closing actuator saturation and positive frequency. The ith disagreement-frequency output satisfies

$$|w_{i,k} - \bar{w}_k| \le W_d, \qquad W_d = \max_i \frac{1}{H}\sqrt{R[\bar{P}^{-1}]_{22}(L^\dagger)_{ii}}. \tag{79}$$

Applying the state bound to y_{k+1} covers w_k; y_1 covers w_0. As $u = w - \delta\omega$, we obtain

$$|u_{i,k}| \le W_c + W_d + \|\delta\omega\|_\infty, \qquad \omega_i(t) \ge \omega_{\mathrm{ref}} - W_c - W_d. \tag{80}$$

First derive these bounds for the unconstrained trajectory. If the first is strictly below $u_{\max}$ and the second is positive, the limiter is never reached, so the constrained execution coincides with that trajectory. This continuation argument closes the assumption of no clipping. The buffer bound must likewise fit within physical capacity; protecting a buffer by blocking execution is a different outcome.

Together, the following sufficient conditions connect the controller and network parameters to the required properties:

$$\begin{aligned}
&V_1 \le R, \qquad R\Psi_i \le r^2 \quad \text{for every } i,\\
&0 < \beta_e^* < C \quad \text{for every } e,\\
&R[\bar{P}^{-1}]_{11} R_e^{\mathrm{eff}} < m_e^2 \quad \text{for every } e,\\
&R[\bar{P}^{-1}]_{22} R_{\max}^{\mathrm{eff}} < H^2\Delta_{\mathrm{safe}}^2,\\
&W_c + W_d + \|\delta\omega\|_\infty < u_{\max},\\
&\omega_{\mathrm{ref}} - W_c - W_d > 0,
\end{aligned} \tag{81}$$

provided the endpoint inequalities of Lemma 4.1 hold. The frequency-spread inequality is required only when a specific threshold is an objective; convergence alone does not require choosing that threshold. All quantities are evaluated for one consistent model, initialization and parameter set. In particular, changing H, h, C, node degree, controller gains or topology can change more than one inequality. These are sufficient certification tests, not a characterization of the largest safe set or necessary stability boundaries.

4.7 The prescribed triangle certificate

Take the three-node complete graph, giving $\mu_2 = \mu_3 = 3$, $R_e^{\mathrm{eff}} = 2/3$ and $(L^\dagger)_{ii} = 2/9$. The parameters are

$$\begin{aligned}
H = h &= \frac{30000000}{29999}, & k_p &= \frac{1}{500000}, & k_i &= \frac{1}{20000000000},\\
K_e &= \frac{1}{10}, & K_{\Delta e} &= \frac{29999}{7500}, & K_u &= -\frac{768}{18749375},\\
C &= 2048, & \beta_e^* &= \beta_e(0) = 1024, & \lambda_e &= 1024.
\end{aligned} \tag{82}$$

Frequencies in this numerical example are normalized, and physical time is measured in reference ticks. Use $\omega^u = (1.0002, 0.9998, 0.9999)^\top$ and its average as $\omega_{\mathrm{ref}} = 29999/30000$, so $\delta\omega = (7, -5, -2)^\top/30000$ and $\bar{w}_0 = 0$. At the compatible startup (56), use

$$\bar{P} = \frac{1}{50000}\begin{bmatrix} 9 & 181 \\ 181 & 49629 \end{bmatrix}, \quad \rho = \frac{1599}{1600}, \quad r = 1. \tag{83}$$

Here $\det \bar{P} = 4139/25000000 > 0$ and the leading diagonal entry is positive. Since the sum of its four entries is one,

$$R_0 = V_1 = H^2\delta\omega^\top L\delta\omega = \frac{234000000}{899940001}. \tag{84}$$

This exact fraction is used in all inequalities. The displayed approximation 0.2600173342 is slightly below the exact value and is not substituted as an upper energy radius.

The sector is $[5/6, 2]$, hence the spectral endpoints are $5/2$ and 6. For $D_\nu = \rho\bar{P} - B(\nu)^\top\bar{P}B(\nu)$, exact rational arithmetic verifies positive leading minor and determinant at both ends:

ν	$[D_\nu]_{11}$ (approx.)	$\det D_\nu$ (approx.)
5/2	$8.2193625299 \times 10^{-7}$	$7.8876135318 \times 10^{-11}$
6	$2.0776400066 \times 10^{-6}$	$7.3380136479 \times 10^{-9}$

The accompanying machine-readable certificate stores the exact fractions; the decimals are only a readable display, not rounded tests of positivity. Using R_0 gives the following conservative, outward bounds:

Quantity	Certified bound
$\sup_{i,k}(\lvert X_{i,k}\rvert + \lvert Y_{i,k}\rvert)$	$< 0.171455 < 1$
Coarse edge excursion	< 32.237442 frames
$\sup_k \lvert\bar{w}_k\rvert$	< 0.003682181
$\sup_{i,k} \lvert w_{i,k} - \bar{w}_k\rvert$	< 0.000250634
$\sup_{i,k} \lvert u_{i,k}\rvert$	$< 0.004166148 < 0.01$
$\inf_{i,t} \omega_i(t)$	> 0.99603385

The coarse excursion already fits within the 1024-frame lower and upper headroom. The frequency bounds use the exact rational-square-root enclosures, not a sum of prematurely rounded display values.

4.8 The exact PI all-time baseline

For $J \in \{\mathrm{PI}, \mathrm{F}\}$ define

$$B_J = \sup_{t\ge 0}\max_e |\beta_e^J(t) - 1024|. \tag{85}$$

The supremum varies time and receiving edges, while the initialization and intrinsic-frequency vector remain fixed. For the same plant, initialization and gains, native PI has $D_\alpha = I_n$. Averaging its increments shows that the mean frequency is preserved. On the triangle, both disagreement modes have eigenvalue 3. Define $\chi_0 = 0$, $\chi_1 = 1$ and

$$\chi_{k+1} = (2 - 3a - 3b)\chi_k - (1 - 3a)\chi_{k-1}. \tag{86}$$

To derive it, use $v_k = x_k - x_{k-1}$ in the PI state recurrence:

$$x_{k+1} = (2I_n - (a+b)L)x_k - (I_n - aL)x_{k-1}.$$

Since $L\delta\omega = 3\delta\omega$ and $x_1 = H\delta\omega$, induction gives $x_k = H\chi_k\delta\omega$. With $\Delta_0 = \max_i \omega_i^u - \min_i \omega_i^u = 1/2500$,

$$\max_e |\beta_e(t_k) - 1024| = H\Delta_0|\chi_k|. \tag{87}$$

This exact rational sequence is enumerated through $N = 200$. Its unique prefix maximum occurs at $k = 111$, with

$$23.40370130812701 \le \max_{0\le k\le N} H\Delta_0|\chi_k| \le 23.40370130812702. \tag{88}$$

It is not yet an all-time assertion until the tail is bounded.

At $N = 200$ put $\delta\chi_N = \chi_N - \chi_{N-1}$, the scalar response difference. The exact state energy is

$$V_N = R_0\{\bar{P}_{11}\chi_N^2 + 2\bar{P}_{12}\chi_N\delta\chi_N + \bar{P}_{22}(\delta\chi_N)^2\}. \tag{89}$$

The common certificate also covers PI's scalar value $\nu = 3$. Thus for all $k \geq N$, the remaining buffer excursion is at most $\sqrt{(2/3)[\bar{P}^{-1}]_{11}V_N}$, with an outward upper bound 17.95805698665804 frames. This is strictly below the attained prefix maximum. ZOH interpolation therefore makes (88) the exact all-physical-time B_{PI} enclosure, reached at time $111H$.

4.9 The Fuzzy interval prefix and contracting tail

A direct floating-point trajectory can locate a candidate peak but does not enclose the exact one. The certificate instead propagates intervals for all components of x_k and v_k on a grid $S^{-1} = 10^{-60}$. A rational number q_{rat} is enclosed by

$$\left[\frac{\lfloor Sq_{\mathrm{rat}}\rfloor}{S}, \frac{\lceil Sq_{\mathrm{rat}}\rceil}{S}\right].$$

Each multiplication uses the extrema of the four endpoint products, rounded outward; division uses a strictly positive enclosed denominator and outward-rounded endpoint quotients. Addition and subtraction use endpoint arithmetic. These operations use integers and rational comparisons, not floating-point rounding assumptions.

For the central formula, $m = \min(|X|, |Y|)$ is enclosed by interval absolute values and endpoint minima. Its interaction term can be written $J = \min(X, Y)$ for two positive inputs, $J = \max(X, Y)$ for two negative inputs and zero otherwise. It is coordinatewise nondecreasing, including across the axes; its lower and upper corner values enclose it on any input rectangle. The positive denominator $1 + 2m/5$ then permits a valid interval evaluation of the exact map. The invariant-domain proof ensures that this central formula is applicable to the exact trajectory.

Propagating the genuine bootstrap state through state $N = 200$ gives a prefix maximum enclosed by

$$[17.314056817962,\ 17.314056817963] \quad \text{frames}. \tag{90}$$

The prefix retains all node coordinates, so common drift is not silently discarded. On the triangle the quadratic energy is equivalently

$$V_k = \sum_{i<j}\{\bar{P}_{11}(x_{i,k} - x_{j,k})^2 + 2\bar{P}_{12}(x_{i,k} - x_{j,k})(v_{i,k} - v_{j,k}) + \bar{P}_{22}(v_{i,k} - v_{j,k})^2\}. \tag{91}$$

Evaluating this expression outward gives $\overline{V}_N \geq V_N$. For all later samples,

$$|\beta_e(t_k) - 1024|^2 \leq \frac{2}{3}[\bar{P}^{-1}]_{11}\rho^{k-N}\overline{V}_N. \tag{92}$$

For either controller, an equivalent tail bound that makes the finite-prefix/infinite-tail decomposition explicit is

$$T_N = \sqrt{\frac{2}{3}[\bar{P}^{-1}]_{11}\overline{V}_N}. \tag{93}$$

For PI one can take $\overline{V}_N = V_N$ exactly; for Fuzzy, $\overline{V}_N$ is an outward interval upper bound. Once T_N is below the certified prefix lower bound, no later sample can exceed the prefix maximum. This closes the infinite time horizon without selecting an arbitrarily long simulation duration.

Table 2: Certified excursions in the prescribed fluid reference (frames).

Controller	Prefix peak enclosure	Tail upper bound
PI	[23.40370130812701, 23.40370130812702]	17.95805698665804
Fuzzy	[17.314056817962, 17.314056817963]	12.29271005372874

An outward upper bound for this whole tail is 12.29271005372874 frames, strictly below the prefix lower bound. Including $x_0 = 0$ and intersample interpolation gives $B_{\mathrm{F}} < 17.315$ on the entire physical time axis. The exact Fuzzy peak is more tightly enclosed by (90); the rounded 17.315 value is used for a simple, conservative comparison.

The enclosures used in the comparison are collected in Table 2. The prefix intervals are outward enclosures of actual peak values; the tail entries are upper bounds. Their different roles are essential for the strict comparison.

4.10 Strict comparison, engineering meaning and scope

Theorem 4.4 (Strict headroom benefit for a prescribed initialization). *For the triangle, parameters and compatible initialization above, initialize the PI sample sum, the Fuzzy accumulator and the stored previous percentage error to zero. Both reference executions keep positive physical frequencies, remain strictly inside the buffer interval* $(0, 2048)$ *and never activate the* 0.01 *correction limiter. Buffer deviations and all pairwise frequency differences converge to zero. Their all-time excursions satisfy*

$$B_{\mathrm{F}} < 17.315 < 23.40370130812701 \leq B_{\mathrm{PI}}. \tag{94}$$

For the minimum lower/upper headroom $M_J = 1024 - B_J$,

$$M_{\mathrm{F}} > 1006.685, \qquad M_{\mathrm{F}} - M_{\mathrm{PI}} > 6.08870130812701 \quad \textit{frames}. \tag{95}$$

Proof. The endpoint inequalities and the radius test certify the invariant nonlinear input region. The buffer, correction and absolute-frequency bounds establish capacity safety, absence of clipping and positive frequencies. The invariant disagreement energy contracts, so the buffer and frequency-difference conclusions follow. PI is included by taking $\alpha_i = 1$, which belongs to the same sector; PI also preserves its initial mean frequency. The certified finite-prefix enclosures and the tail bounds imply that both all-time excursions are attained in the prefix. The Fuzzy prefix upper bound is below 17.315, whereas PI attains the enclosed peak at sample 111. Intersample affine occupancy rules out a missed larger fluid peak. Finally, each edge has a reverse edge with the negative occupancy deviation, so the worst lower/upper headroom is exactly $1024 - B_J$. Subtracting the strict excursion bounds gives the stated gain in headroom. □

The comparison uses a certified Fuzzy upper bound and an attained PI peak. Merely comparing two conservative upper bounds would not establish the strict ordering of the actual excursions.

Both reserves are already large. The comparison establishes a conditional resource-margin benefit, not a demonstrated storage bottleneck or hardware cost reduction at this operating point.

The supremum is over time and edges for one prescribed intrinsic-frequency vector. It is not an expanded safe set for all 400-ppm patterns, a guarantee against arbitrary latency perturbations or superiority over independently tuned PI. For a different capacity, $\gamma_i = 100/(d_i C)$ changes; keeping the Fuzzy gains unchanged generally changes the token-domain nonlinear map. A new capacity certificate requires a fresh derivation or a verified map-preserving normalization change. Physical occupancy constraints also do not replace the delayed runtime's in-flight and credit/readiness rules.

This is a computer-assisted mathematical certificate with reproducible integer/rational checks. It is not proof-assistant verification of the Julia implementation. The actual asynchronous, delayed, quantized cases belong in the experimental section. This division preserves the distinction between a theorem about the declared reference and empirical evidence about Callisto.

5 Evaluation

The evaluation connects the controller analysis to three distinct questions: whether the continuous-clock implementation respects the PI calibration; whether the nonlinear reference trajectories agree with the mathematical analysis; and whether the proposed controller offers a measurable benefit against independently tuned PI under shared delivery disturbances. We separate matched-gain mechanism checks from equal-budget controller selection, and reference-model certificates from actual Callisto measurements.

5.1 Experimental design and continuous-clock execution

All actual experiments use the continuous-phase Callisto realization of Section 3. Oscillator phases evolve continuously, integer frame crossings determine buffer events, and measurements and correction applications are scheduled by each node's local phase. There is no common physical polling grid. Table 3 gives the nominal plant; the startup stress study changes the intrinsic frequencies and reconstructs the associated negative-time histories explicitly.

Evidence groups and data separation. The implementation check, M0, contains six matched pairs, or 12 actual runs. An earlier fixed-preset burst campaign, M2, supplies a separate mechanism diagnostic. The earlier fixed-preset random campaign, M3, is development evidence and supplies no seeds to the new independent tests. Its fixed-preset comparison did not support lower Fuzzy RMS against moderate PI; the new study asks the different question of independently selected controllers under an equal evaluation budget. The new training study charges 192 slots, including four verified cached M2 evaluations and 188 new actual runs. Selection is frozen before 186 new held-out runs. A separately registered slower-disturbance study adds 107 actual runs, including its engineering pilots, and the startup study adds six. Its 32 synchronous fluid-reference trajectories are reference calculations, not additional actual runs. Old data, infeasible candidates and repair records are retained in the evidence archive.

Shared delivery inputs. All controllers receive identical externally generated desired-latency profiles within each comparison, verified by content hashes. Delay is 20 before onset, including the negative-time prehistory. Ordered arrivals use the causal running-maximum envelope of Section 3; a delay decrease may release an ordered batch. The model does not impose a finite serialization rate. For bounded random latency, the desired delay is $20 + z_m$, where

$$z_0 = 5\varepsilon_0,$$
$$z_m = r_z z_{m-1} + (1 - r_z)5\varepsilon_m, \quad r_z = e^{-1}, \quad \varepsilon_m \sim \mathcal{U}[-1, 1]. \tag{96}$$

Thus desired delay remains in $[15, 25]$. The index m counts physical profile updates on a grid of spacing T_c, independently of controller polling. Here T_c denotes a profile-update interval, not an exactly measured correlation time. The original two regimes use $T_c = 1000$ and 10000; the later confirmation study uses 100000.

5.2 Metrics, admissibility and comparison windows

For n nodes, let $\bar{\omega}(t) = n^{-1}\sum_i \omega_i(t)$ and define the instantaneous across-node root-mean-square (RMS) disagreement as

$$\sigma_\omega(t) = \sqrt{n^{-1}\sum_i [\omega_i(t) - \bar{\omega}(t)]^2},$$
$$\sigma_{\mathrm{ppm}}(t) = 10^6 \sigma_\omega(t)/\omega_{\mathrm{unit}}, \qquad \omega_{\mathrm{unit}} = 1. \tag{97}$$

Parts per million (ppm) use the fixed normalized frequency unit, not ω_{base}. For a physical-time window $W = [a, b)$ of duration $T_W = b - a$, report

$$R_W = \sqrt{T_W^{-1}\int_W \sigma_{\mathrm{ppm}}(t)^2\, dt},$$
$$I_W = \int_W \sigma_{\mathrm{ppm}}(t)\, dt, \qquad P_W = \sup_{t\in W} \sigma_{\mathrm{ppm}}(t). \tag{98}$$

The integral absolute error (IAE) I_W integrates RMS amplitude, rather than the mean absolute node deviation. Frequency moments use complete held-frequency intervals weighted by their physical durations. The actual pairwise spread $\Delta_\omega = \max_i \omega_i - \min_i \omega_i$ is recorded separately. Although $\Delta_\omega \le \sqrt{2n}\,\sigma_\omega$ pointwise, window RMS alone does not establish a maximum transient-spread bound.

Buffer excursion, minimum two-sided headroom and mean command total variation (TV) are

$$B_W^{\mathrm{obs}} = \max_e \sup_{t\in[a,b]} |\beta_e(t) - \beta_e^*|,$$
$$H_W^{\mathrm{obs}} = \min_e \inf_{t\in[a,b]} \min\{\beta_e(t), C - \beta_e(t)\}, \tag{99}$$
$$\mathrm{TV}_W = n^{-1}\sum_i \sum_{k:\, a\le t_{i,k}^a \le b} |u_{i,k} - u_{i,k-1}|. \tag{100}$$

Buffer statistics reconstruct all integer crossings and ordered arrivals, grouping simultaneous events and including the endpoint state. Sparse plot samples do not determine extrema. TV includes both endpoints under the frozen convention, so adjacent-window activities cannot be added without treating the shared endpoint. TV measures command activity, not hardware energy. Requested and applied corrections, saturation, gate events, positive frequency and data-phase progress are checked separately.

Table 3: Nominal actual-Callisto configuration. Time is in physical reference ticks except where local ticks are explicitly stated. Controller parameters selected for independent tests are listed separately in Table 5.

Parameter	Value
Topology	Three-node bidirectional triangle; six receiving buffers
Intrinsic frequencies ω^u	$(1.0002, 0.9998, 0.9999)^\top$; initial spread 400 ppm
Initial phase; base frequency	0.1 at every node; $\omega_{\text{base}} = 29999/30000$
Capacity; initial occupancy; target	$C = 2048$; $\beta_e(0) = \beta_e^* = 1024$ frames
Initial in-flight frames; marking	20 per channel; 1044 in the recorded convention
Nominal path delay	20
Polling; control-execution delay	$p = 1000$; $d_{\text{ctrl}} = 10$ local ticks
Execution constraints	$\lvert u_i \rvert \le 0.01$; guards enabled; reserve zero; no anti-windup
Control law	Native integrate-first PI or the complete incremental 25-rule Fuzzy law
Other mechanisms	Observer and supervisory mode switching disabled
Disturbance onset	$t_{\text{on}} = 2 \times 10^6$

Before each post-centering disturbance, three consecutive 10^5-tick windows ending at onset must satisfy disagreement RMS at most 2 ppm, each buffer's absolute mean bias at most two frames and standard deviation at most one frame, common-frequency temporal standard deviation at most 1 ppm and three-window mean range at most 1 ppm, with positive progress and no intervention. These are finite engineering centering criteria, not exact equilibrium. A training candidate failing the same criteria in any required case remains infeasible and consumes its assigned budget.

The original-regime tests use $W = [2 \times 10^6, 4 \times 10^6)$ and tail $[3.6{\times}10^6, 4{\times}10^6)$. Each controller's own clean trajectory is its same-parameter, nominal-delay counterfactual over the same physical times. Common-frequency shift is reported relative to this counterfactual, separately from relative synchronization. Strict recovery requires the centering criteria for three consecutive windows and a window-mean common frequency within 2 ppm of own clean. A return not observed by the specified endpoint is right-censored, not assigned a recovery time of zero or interpreted as permanent nonrecovery.

5.3 Implementation and nonlinear-reference validation

M0 checks Proposition 3.1 by replacing the Fuzzy surface with its calibrated linear counterpart F_0, without changing the continuous plant. Six pairs cover polling periods 100 and 1000, initial frequency spreads 35 and 400 ppm, and capacity/target settings $(2048, 1024)$, $(512, 128)$ and $(512, 384)$ over 2×10^5 ticks. Across 19,800 paired updates, the maximum correction difference is 5.29×10^{-18}; physical/data phases, measurement vectors and saved buffer counts agree, without saturation or guard intervention. This checks initialization, integration order and gain mapping. It does not replace the complete nonlinear law in the performance experiments.

The original 25-rule law is also evaluated directly on the synchronous, zero-delay reference of Section 4, with 2000 post-bootstrap updates. At the prescribed moderate-gain initialization, the fluid buffer peak is approximately 17.314056818 frames, maximum $|X| + |Y|$ is 0.090130790, maximum absolute correction is $3.21398377{\times}10^{-4}$, and no contraction-envelope violation is observed. Physical frequencies stay in $[0.9998, 1.0002]$ including startup. These finite numerical checks agree with the analysis; the all-time comparison, Fuzzy below 17.315 versus PI approximately 23.403701 frames, follows from the interval-prefix and contracting-tail proof in Theorem 4.4.

Rule verification. We reconstruct requested corrections from the saved inputs in all 299 new validation, slow-delivery and actual startup records. The maximum residual is approximately 5.42×10^{-20}. This uses the complete rule map and stored accumulator, not a sector surrogate. The earlier M2 diagnostic additionally stays within Ω_5, with observed $\max(|X| + |Y|) = 2.001888$, supporting application of Proposition 3.2 to those measured inputs. This diagnostic is retained separately from the independently selected controllers: Fuzzy_B reaches approximately 39.159 in the large held-out burst. The full 25-rule law remains defined there, whereas the central-region identity and the Ω_1 reference witness do not cover that whole trace. Same-input memory accounting is an algebraic diagnostic, not an additional closed-loop F_0 experiment.

5.4 Reference safety strata and actual startup transients

To vary initial mismatch without changing occupancy normalization, capacity and targets remain fixed while

$$\omega^u(s) = \frac{29999}{30000}\mathbf{1} + s\frac{(7, -5, -2)^\top}{30000}, \quad s \in \{0.5, 1, 2, 4, 8, 16, 32, 64\}. \tag{101}$$

Here s is an initial-mismatch scale, with intrinsic spread $400s$ ppm. The reference uses the matched moderate coordinates $(k_p, k_i) = (2{\times}10^{-6}, 5{\times}10^{-11})$, synchronous zero-delay fluid execution and $H = h = 30000000/29999$. Its 32 trajectories comprise two controllers, eight scales and both unprotected virtual and 0.01-clipped variants. Each includes the bootstrap and 2000 subsequent zero-order-hold intervals. The initial certificate energy scales as $V_1(s) = s^2 V_1(1)$; input-domain, buffer, positive-frequency and correction conditions are checked jointly.

The largest grid point satisfying the common sufficient conditions is $s = 2$. At $s = 4$, both reference trajectories remain normal over the record although the certificate no longer guarantees them. At $s = 32$, unprotected reference requests first require actuator

Table 4: Actual startup stress at matched moderate coordinates. Buffer excursions include startup and all reconstructed events. Sat. counts saturated control updates; every case has zero gate events and replay mismatches.

		Buffer excursion / frames		Sat.
s	Spread / ppm	PI	Fuzzy	PI / Fuzzy
2	800	47	37	0 / 0
4	1600	94	72	0 / 0
32	12800	750	664	176 / 85

clipping; at $s = 64$, both virtual reference buffer trajectories violate capacity. Figure 1 distinguishes measured virtual peaks from sufficient bounds. Failure of a sufficient condition is inconclusive, rather than a prediction of instability.

Before running actual cases, the selection rule freezes three common scales: 2, 4 and 32, representing certified, observed-safe uncertified, and correction-constrained reference strata. Actual Callisto restores path delay 20 and local polling/control delay 1000/10. Its locked startup horizon is $2001H \simeq 2.001067 \times 10^6$ physical ticks, rather than the 4×10^6 used for the disturbance tests. This shorter horizon is an execution-plan deviation from the earlier completion instruction and limits the empirical safety observation accordingly. The six cases are startup stress tests, not post-centering disturbance tests.

Fuzzy has a smaller maximum actual buffer excursion at all three selected scales (Table 4). At $s = 4$, the reduction is from 94 to 72 frames, approximately 23.4%, with no saturation or gate intervention. Figure 2 shows the associated frequency and occupancy transients. All six records retain positive frequency, data progress and capacity compliance. At $s = 32$, Fuzzy also requests fewer saturated updates, but 12800 ppm is a stress setting, not a typical hardware tolerance. Its histories are initialized consistently: the faster source has 21 in-flight frames and marking 1045, whereas the other sources retain 20 and 1044. Both controllers receive identical histories at each scale.

These observations show a smaller transient buffer demand at matched gains, but do not transfer the reference theorem to delayed, asynchronous, quantized execution. The recorded pairwise spread remains a separate quantity. No hardware-based numerical Δ_{safe} is assumed; applying the reference spread corollary would require an independently justified threshold exceeding $D(R)$.

5.5 Equal-budget selection and independent random-latency tests

Fixed search budget and selection. Each family uses the same 24-point grid,

$$k_p \in \{1.5, 2, 2.5, 3, 4, 6\} \times 10^{-6},$$
$$k_i \in \{1, 2, 5, 10\} \times 10^{-11}. \tag{102}$$

Fuzzy fixes $K_e = 0.1$ and obtains $K_{\Delta e}, K_u$ from the calibration in Section 3; it receives no extra shape-search dimension. Four training cases per candidate comprise clean operation, a single-edge 20-tick delay increment lasting 10^5 ticks, and two random regimes using training seed 31001. Both families therefore receive 96 evaluation slots. Four hash-verified cached evaluations remain charged within the 192-slot total. Nineteen PI and 22 Fuzzy candidates satisfy the same finite-run admissibility requirements; all seven infeasible candidates are retained without replacement.

Five training objectives are minimized separately: worst random-case RMS, burst IAE, worst disturbance-window buffer excursion, peak absolute common-frequency excursion from own clean, and worst mean command TV. Each family's feasible Pareto set supplies a random-priority representative, labeled R, and a burst-priority representative, labeled B. The frozen lexicographic rule prioritizes random RMS for R and burst IAE for B, with the remaining objectives and candidate identifier as declared tie breakers. One PI serves both priorities; Fuzzy selects distinct representatives (Table 5). Deduplication gives three candidates, without backfilling.

Untouched paired tests. Only after freezing selection are test profiles generated from seeds 41001–41030. All three candidates execute both original random regimes and both held-out burst combinations, giving $30 \times 2 \times 3 + 2 \times 3 = 186$ new actual runs. All 186 complete and satisfy the declared normal-admissibility checks, with no engineering QA errors or complete-event replay mismatches. No seed pair is excluded. Archived M3 seeds are not reused as untouched tests.

The primary random comparison is Fuzzy_R versus PI_R. Let ℓ index independent seeds. Report the paired mean difference and mean of per-seed ratios,

$$\widehat{d} = \frac{1}{30} \sum_{\ell=1}^{30} (R_{W,\ell}^{\text{Fuzzy}_\text{R}} - R_{W,\ell}^{\text{PI}_\text{R}}),$$
$$\widehat{r} = \frac{1}{30} \sum_{\ell=1}^{30} R_{W,\ell}^{\text{Fuzzy}_\text{R}} / R_{W,\ell}^{\text{PI}_\text{R}}. \tag{103}$$

The ratio is not the ratio of group means. Confidence intervals (CIs) use 20000 paired-seed percentile-bootstrap resamples, with fixed resampling seeds 58001/58002. Pointwise 95% intervals accompany each endpoint; a joint statement about the two original regimes also uses nominal Bonferroni-adjusted 97.5% intervals per regime. Time points, edges and bootstrap draws are not independent replications.

Table 6 and Figure 3 show reductions in disagreement RMS at both tested profile grids. The mean paired relative improvements are 16.78% and 7.85%. The nominal family-adjusted difference intervals are $[-0.81274, -0.73483]$ and $[-0.44451, -0.28407]$ ppm, also below zero. This supports an advantage of the selected Fuzzy scheme under the declared finite-grid budget, plant and two input regimes. Because the selected coordinates differ between families, it does not isolate a same-gain inference-only effect or establish globally optimal PI/Fuzzy performance.

Activity and common-frequency cost. Mean per-node command TV decreases from 0.0102361 to 0.00782282 at $T_c = 1000$, and from 0.00112361 to 0.000906694 at $T_c = 10000$ (PI_R to Fuzzy_R). All selected candidates have at most six frames of disturbance-window target deviation. However, Fuzzy_R has larger magnitude mean tail common-frequency shifts from own clean: -61.420 versus -15.692 ppm, and -31.933 versus -15.384 ppm. Thus reduced relative disagreement and command activity do not imply improved absolute-frequency regulation. No absolute-frequency reference loop is present.

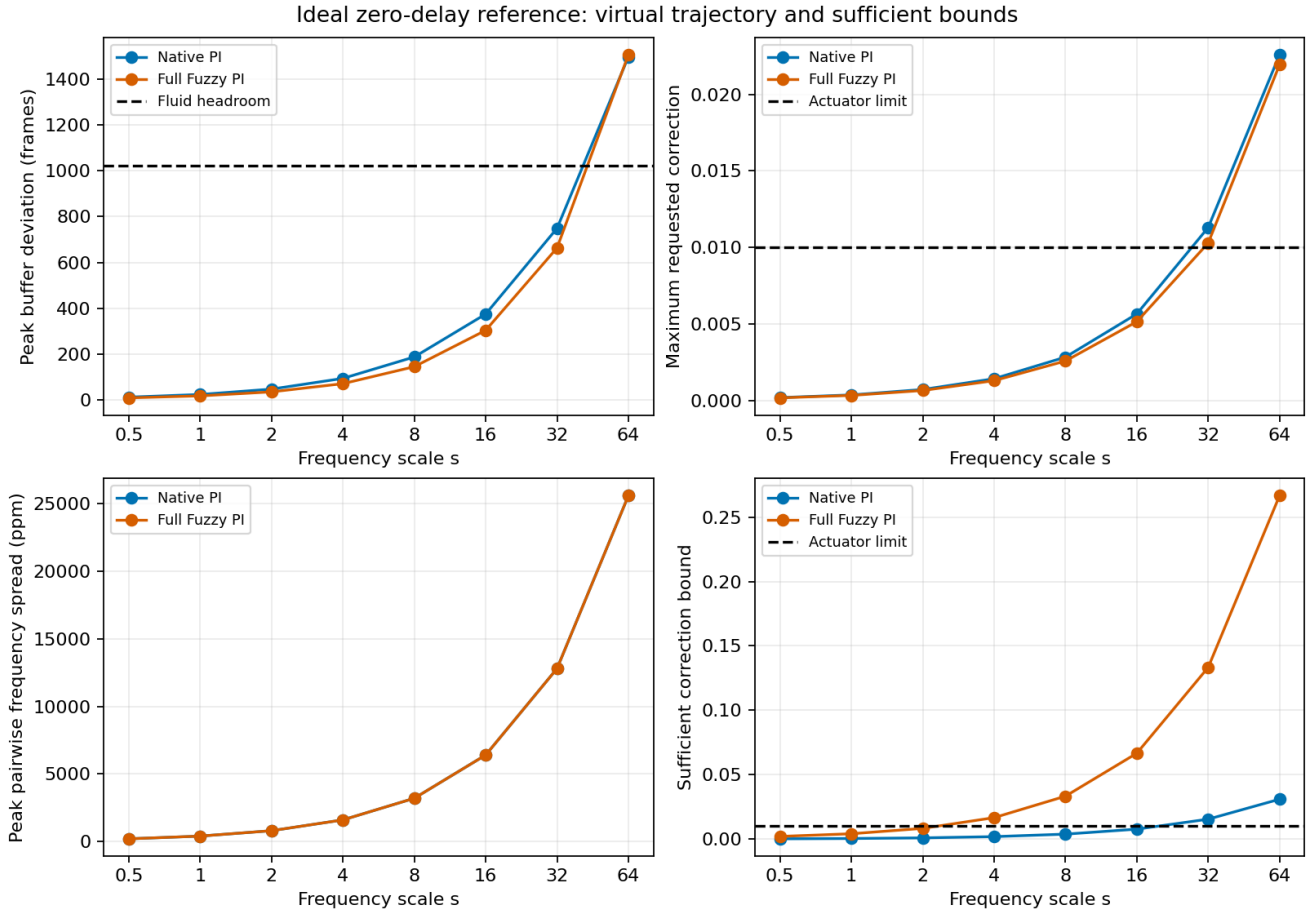


Figure 1: Synchronous zero-delay fluid-reference sweep at matched moderate coordinates. Virtual trajectory peaks and analytical sufficient bounds are distinct quantities. The dashed levels show fluid buffer headroom and the correction limit; clipping and unprotected virtual trajectories are archived separately. These are reference calculations, not actual Callisto safety certificates.

Table 5: Frozen representatives. The k_p, k_i columns are native PI gains for PI and calibrated PI coordinates for Fuzzy. R and B denote random- and burst-priority selection; the candidates are not retuned on tests.

Representative	Candidate	k_p	k_i	K_e	$K_{\Delta e}$	K_u
$\mathrm{PI}_R = \mathrm{PI}_B$	PI_g05	2×10^{-6}	10^{-11}	—	—	—
Fuzzy_R	Fuzzy_g02	1.5×10^{-6}	2×10^{-11}	0.1	7.499750	-1.638455×10^{-5}
Fuzzy_B	Fuzzy_g17	4×10^{-6}	10^{-11}	0.1	39.998667	-8.192273×10^{-6}

Table 6: Untouched original-regime tests: 30/30 admissible pairs per row. Controller RMS entries are seed means in ppm. Difference is Fuzzy_R minus PI_R; ratios are means of paired ratios. Brackets give pointwise 95% paired bootstrap intervals.

Profile grid	PI_R RMS	Fuzzy_R RMS	Difference / ppm [CI]	Paired ratio [CI]
$T_c = 1000$	4.61355	3.83949	−0.77406 [−0.80767,−0.73992]	0.83217 [0.82484,0.83964]
$T_c = 10000$	4.65204	4.28718	−0.36486 [−0.43523,−0.29464]	0.92150 [0.90630,0.93672]

5.6 Held-out bursts and slower delivery changes

Burst-priority validation. The two predetermined held-out combinations are a 10-tick desired delay increment lasting 10^4 ticks

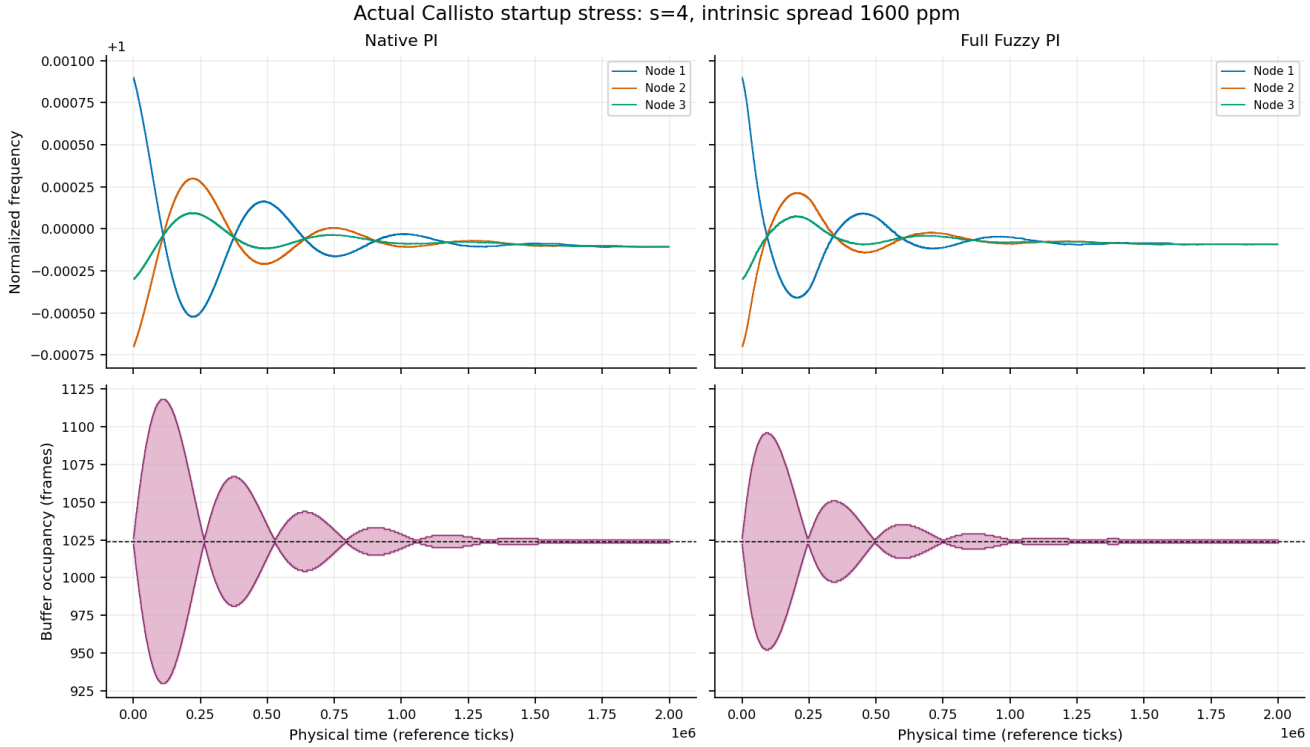


Figure 2: Actual continuous-phase Callisto startup at $s = 4$ (1600 ppm), with matched moderate coordinates. Upper panels show positive absolute normalized frequencies using a $+1$ axis offset. Lower panels show the minimum–maximum occupancy envelope across all receiving buffers, from the complete crossing stream. Fuzzy reduces the maximum target deviation from 94 to 72 frames over the stated startup horizon.

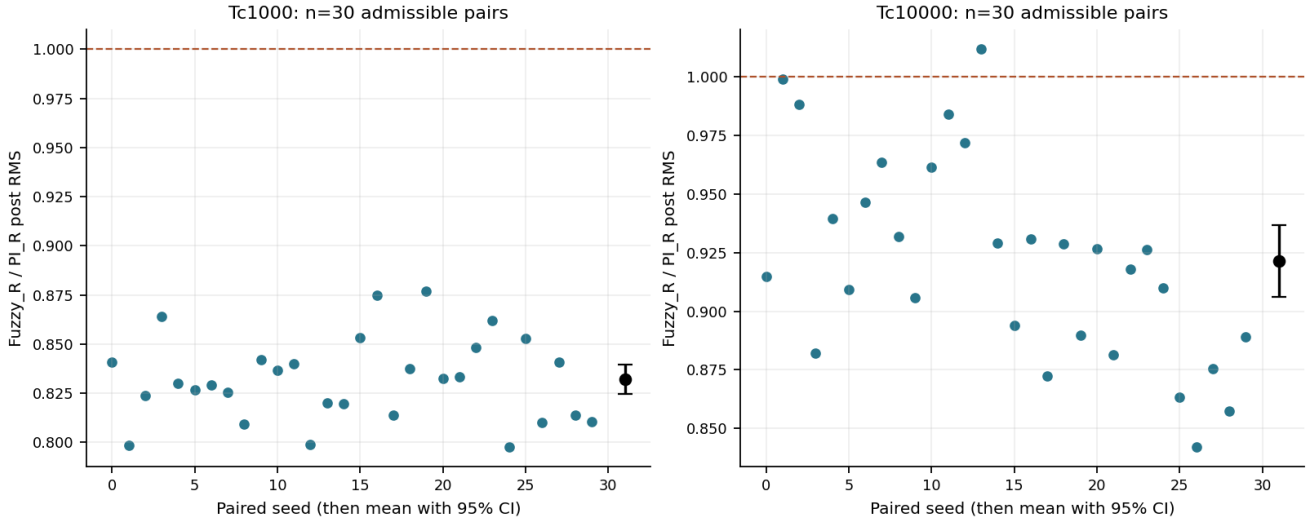


Figure 3: Independent paired $\text{Fuzzy}_R/\text{PI}_R$ RMS effects after equal-budget selection. Each dot is one of 30 untouched seed pairs. The final black marker is the mean paired ratio with a pointwise 95% CI; the dashed line denotes equal RMS. Individual seed outcomes and their aggregate are both retained.

on channel $2 \longrightarrow 3$, and a 40-tick increment lasting 10^5 ticks on $3 \longrightarrow 1$, both beginning at 2×10^6. They are distinct from the training combination. For the larger burst, Fuzzy_B lowers post-onset

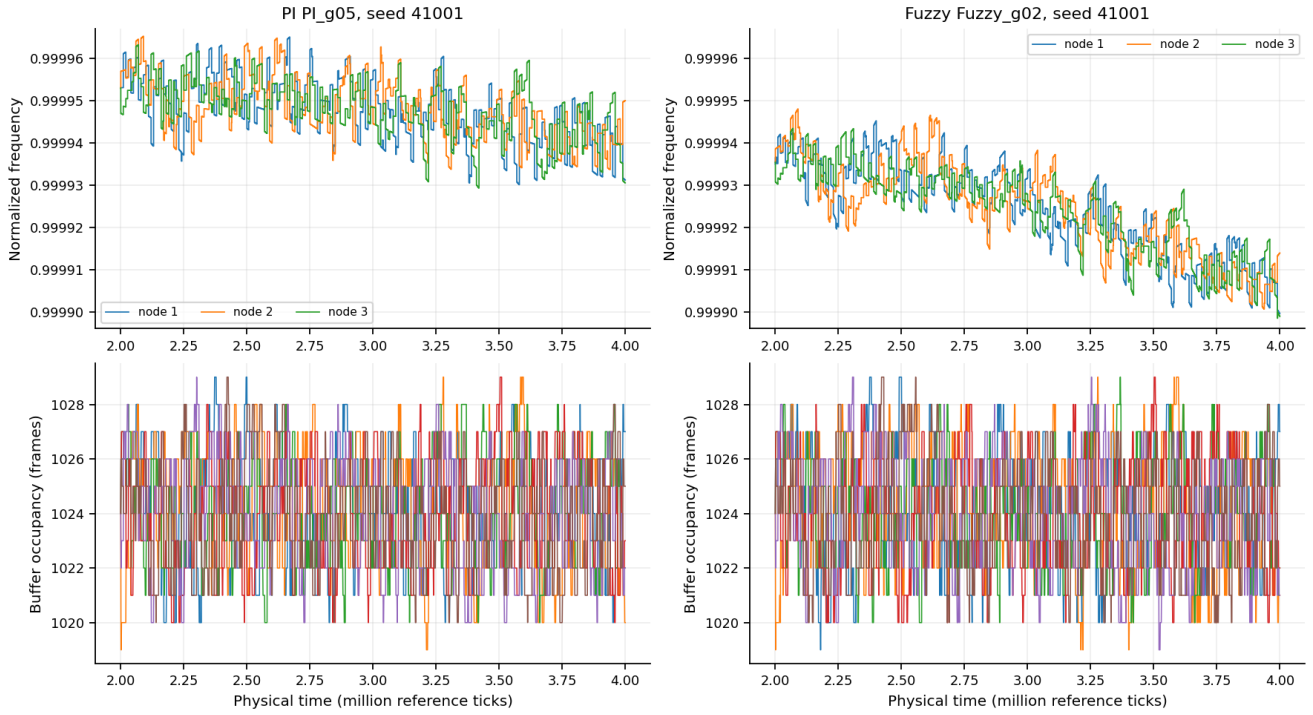


Figure 4: Saved raw traces for the first held-out seed (41001) in the $T_c = 10000$ random experiment. Frequency uses held corrections and shared physical time; buffer occupancy overview samples are for display. Reported moments and buffer extrema instead use complete intervals and events. The paired-seed statistics, rather than this one trace, support the mean performance comparison.

Table 7: Both held-out burst-priority cases over $[2\times10^6, 4\times10^6)$. IAE is in 10^6 ppm·ticks; buffer excursion is in frames. A and D denote desired delay increment and its duration, respectively.

	IAE		Buffer excursion	
Case	PI_B	$Fuzzy_B$	PI_B	$Fuzzy_B$
$A = 10, D = 10^4$	0.265814	0.352614	11	11
$A = 40, D = 10^5$	8.192410	3.623166	40	40

IAE from 8.19241×10^6 to 3.62317×10^6 ppm·ticks, approximately 55.8%, while matching the 40-frame buffer excursion. Table 7 retains both deterministic cases: the improvement depends on the burst combination rather than holding for every tested burst. These two cases do not supply seed-level CIs.

Ramp–hold–ramp inputs. A separate registered study freezes the selected gains and applies a 20-tick delay increment on channel $1 \to 2$ using rise/fall times $T_r \in \{10^4, 10^5\}$ and hold times $T_h \in \{10^5, 10^6\}$. The four combinations all run to 5.2×10^6 ticks. The end of the falling ramp is $t_{\text{off}} = 2 \times 10^6 + 2T_r + T_h$; each case has its own $[t_{\text{off}}, t_{\text{off}} + 2 \times 10^6)$ recovery window and same-horizon clean counterfactual. A larger constant delay plateau does not mean delivery stops throughout that plateau. The ramp instead gives feedback time to act while the delivery imbalance forms.

For $T_r = 10^5$, maximum buffer deviations are 19 frames for PI, 18 for $Fuzzy_R$ and 17 for $Fuzzy_B$. The slower formation of the disturbance therefore provides another observed buffer-margin benefit. Frequency RMS favors PI across the four ramp combinations, and strict own-clean recovery is censored in all 12 ramp records. Different disturbance durations are not pooled into a raw-IAE ranking; these data support a resource/frequency tradeoff, not a faster-settling claim.

Separately registered slow-random confirmation. The slower grid $T_c = 100000$ uses new seeds 51001–51030, unchanged selected gains, the same bounded recursion, and $W = [2 \times 10^6, 5 \times 10^6)$ with tail $[4.6 \times 10^6, 5 \times 10^6)$. Each trace has 30 complete profile blocks, which are not independent statistical replicates. The study includes 105 matrix slots: three long-clean cases, 12 ramps and 90 random cases. Four matrix slots reuse qualified pilot runs; 101 new main runs plus six pilots give 107 unique actual runs. All matrix slots pass profile, nominal-prefix and event-replay checks, with no excluded confirmation seed pairs.

This confirmation bounds the applicability of the original-regime advantage: the mean paired $Fuzzy_R/PI_R$ RMS ratio is 1.33312 with pointwise 95% CI [1.30337,1.36219], using 20000 paired bootstrap resamples and fixed seed 59001. The mean paired difference is +1.53215 ppm [1.39578,1.66175]. Slower variation alone therefore

does not extend the selected controller's RMS advantage. This confirmation delimits the two original-regime benefits without changing their registered primary endpoints or the frozen controller selection.

5.7 Connection between analysis and measured benefits

The evidence supports two benefits. First, nonlinear analysis at matched gain coordinates proves a strict headroom improvement for the prescribed fluid reference; actual matched-gain startup tests also show smaller measured excursions at three fixed initial mismatches. Second, equal-budget, independent selection yields lower disagreement RMS and mean command activity in two held-out random regimes, while the burst-priority representative reduces IAE in the larger held-out burst. These are distinct statements with different gains, assumptions and observation windows.

Their scope is explicit: common-frequency shift remains a cost, the small burst and slower delivery tests delimit performance, and finite actual records do not certify all-time safety or convergence. The experiments do not establish a general safe-set inclusion, a globally optimal comparator, payload-level functional logical synchrony, or hardware energy savings. Positive phase progress and buffer compliance show normal model execution; they are not a functional refinement proof. Within these limits, the combined analysis and tests provide both a mechanistic reason and independent empirical evidence for considering the proposed Fuzzy PI law when transient buffer use and disturbance-dependent synchronization performance matter.

6 Discussion and Limitations

The results identify two reasons to consider the proposed Fuzzy PI law for bittide: a provable reduction in transient buffer demand for a specified reference execution, and independently tested synchronization benefits in defined delivery regimes. The connection between these results is the explicit nonlinear correction mechanism, rather than an assumption that fuzzy inference improves every performance metric. This section explains how the benefits should inform controller selection and which extensions require further evidence.

6.1 What nonlinear response shaping adds

The controller offers a local way to change the relationship between occupancy error, its change between measurements and the next frequency correction. Proposition 3.2 shows that the normalized min-activation surface strengthens its calibrated linear counterpart for aligned input signs and attenuates its magnitude for opposing signs within the stated central region. A uniform increase in PI gains would scale both cases uniformly. The Fuzzy law therefore supplies a different response shape at calibrated gain coordinates, while retaining a concrete connection to the native PI implementation.

This interpretation concerns a correction increment, not a complete closed-loop performance ordering. In particular, aligned signs do not always mean that absolute occupancy error is increasing; an error can cross zero while decreasing in magnitude. The plant, measurement history and accumulated correction determine the resulting trajectory. Equation (47) makes the memory contribution explicit: past nonlinear increments may remain in the accumulator even when later inputs have zero nonlinear residual. This provides a mechanism to investigate transient shaping and persistent offsets, but a same-input decomposition alone does not establish their causal contribution to differences between independently evolving closed loops.

The comparisons address two questions. The reference proof and startup tests hold the calibrated gain coordinates fixed. They test a prescribed change in the controller law. The equal-budget study instead asks whether each controller family can produce a useful selected design with the same finite search budget. Its lower held-out RMS supports the selected Fuzzy scheme, including its chosen gains, rather than an inference-only effect at identical gains. Keeping these questions separate makes the empirical benefit more meaningful than a comparison against a single arbitrary PI setting.

6.2 Engineering meaning of the buffer-headroom benefit

For a finite receiving buffer, the largest transient deviation matters even when the eventual operating point is admissible. Theorem 4.4 compares a certified upper bound for the Fuzzy excursion with an attained PI peak, not just two conservative upper bounds. For its prescribed initialization, the resulting improvement in minimum headroom exceeds 6.0887 fluid frames. Both controllers also satisfy the reference model's positive-frequency, correction and convergence conditions. The benefit thus concerns transient resource use while those conditions remain satisfied.

The matched-gain actual tests provide a separate engineering observation. For example, at 1600 ppm, Fuzzy reduces the measured excursion from 94 to 72 frames without correction saturation or gate intervention (Table 4). This is consistent with a useful headroom effect after introducing delayed local execution and integer buffer events, although the mathematical reference certificate does not cover that implementation. At the larger stress scale, fewer saturated updates and smaller excursions remain finite-run findings, not a proof that saturation is harmless or that future overflow is impossible.

These results measure reserve at fixed capacity. They do not establish that a smaller hardware buffer can simply be substituted. Capacity appears in the feedback normalization $\gamma_i = 100/(d_i C)$, where d_i is the number of receiving neighbours. Changing C or the degree changes the token-domain controller map unless the gains or normalization are adjusted consistently. Any capacity-reduction claim would therefore require a new model-consistent comparison and a fresh safety check. Similarly, a failed sufficient condition is inconclusive: the observed-safe but uncertified reference cases illustrate conservatism, not an enlarged general safe set.

6.3 Selecting a controller for the operating regime

The independent tests demonstrate a practical benefit under the stated selection policy. With identical evaluation budgets, frozen gains and shared untouched profiles, the random-priority Fuzzy representative reduces disagreement RMS and mean command TV in both original random regimes. The burst-priority representative also substantially reduces IAE in the larger held-out burst. The

distinct selected representatives show that the desired task matters: prioritizing random disagreement and prioritizing a burst transient need not choose the same parameters. The experiments do not switch between these representatives online.

Several measures are needed to interpret the benefit. Relative frequency agreement concerns differences between nodes; it does not require the common frequency to equal one or to return to its clean counterfactual. The larger common-frequency shift accompanying the selected Fuzzy RMS improvement is therefore an engineering cost to evaluate against the application, not an inconsistency in the RMS measurement. The controller has no absolute-frequency reference loop. Likewise, smaller buffer peaks need not coincide with the lowest RMS, and command TV is an activity metric rather than a measured energy cost. A design choice should specify the acceptable headroom, frequency-shift and correction constraints before selecting among the reported tradeoffs.

The slower-delivery study further shows why timing conditions belong in that choice. Increasing a constant path-delay plateau does not suspend arrivals for the whole plateau: its onset changes in-flight inventory, and the first deficit can form before changed feedback is applied. The ramp profiles allow regulation during the formation of the imbalance, whereas the random profile grids change its temporal structure. Neither a longer hold nor slower variation alone establishes an RMS advantage. These observations motivate matching the controller to a declared disturbance regime; they do not identify an optimal polling period, since polling was not independently swept in the new performance tests.

6.4 Meaning of safety, convergence and logical synchrony

The certificate applies to the synchronous zero-delay fluid model with its compatible targets, initialization and invariant input region. Its sampled recurrence is the exact endpoint map of that continuous zero-order-hold reference. It is not a replacement discrete firing-count plant, nor an exact model of the asynchronous controller scheduling and integer delivery events in actual Callisto. In particular, the complete 25-rule controller remains executable outside the central input region, while the regional formula and the numerical invariant certificate cannot be applied to arbitrary such trajectories.

The actual tests establish capacity compliance, positive frequency and progress over specified finite records. Persistent random input is evaluated by windowed disagreement and occupancy statistics rather than claimed exact convergence to a fixed equilibrium. The startup records also have a shorter declared horizon than the disturbance tests. None of these observations alone supplies an all-time safety or convergence proof for delayed execution. Similarly, the reference frequency-spread result requires a separately justified engineering threshold; logical synchrony does not supply a universal ppm limit.

Logical synchrony is the computational requirement motivating the controller, while physical syntony, buffer centering and positive frequency describe different properties of its implementation. Preserving capacity and frame order, together with continued data progress, supports the normal execution assumed here. Demonstrating functional logical synchrony for an application would additionally require checking the logical frame relationships and payload execution against the corresponding logical model. These experiments do not perform that refinement check or measure application-level useful throughput. A capacity guard that prevents underflow by stopping data progress must also be distinguished from successful unstalled control.

6.5 External validity and next steps

The independent comparison is deliberately bounded: a three-node triangle, a fixed two-dimensional gain grid, fixed Fuzzy input scaling K_e, and one training seed per random regime. The untouched seeds quantify variation within those declared input families; they do not turn the search into global controller optimization or establish performance on larger, heterogeneous or changing networks. The transport model preserves order and permits batch release without a finite serialization rate. Hardware oscillator limits, serialized links, packet loss, node dropout and controller execution cost require separate evaluation. The current observer and supervisory switching mechanisms are disabled, so they contribute no fault-recovery claim.

The most direct theoretical extension is to account explicitly for local sampling, bounded delay and quantization when constructing invariant buffer and input regions. Such a result should preserve absolute-frequency and actuator constraints as well as disagreement bounds. Empirically, broader topologies and delivery models would test whether the observed benefits persist beyond the present triangle. If absolute-rate requirements or sustained saturation are important, a common-frequency constraint or anti-windup mechanism could be studied as a separate controller extension, followed by fresh analysis and independent tests. These steps extend the present mechanism and evidence without changing the scope of its proved or measured benefits.

7 Conclusion

We proposed a decentralized incremental Fuzzy PI controller for bittide and connected its complete rule law to native PI through explicit, implementation-checked calibration. The exact central nonlinear map explains how the controller reshapes correction increments according to the relationship between occupancy error and its change. For the stated synchronous zero-delay fluid reference, invariant-domain and Lyapunov analysis establish convergence of buffer deviations and pairwise frequency differences, with positive physical frequencies and the stated safety and execution constraints. A certified trajectory prefix and contracting tail prove a strict headroom benefit for one prescribed matched-gain initialization: Fuzzy excursion is below 17.315 frames, while PI attains approximately 23.403701 frames.

Continuous-phase Callisto tests separately evaluate delayed local execution. Matched-gain startup experiments show smaller measured buffer excursions at three prescribed initial mismatches. After equal-budget tuning and frozen selection, 186 untouched tests give mean paired RMS improvements of 16.78% and 7.85% for the selected random-priority Fuzzy controller in two bounded-random

regimes, alongside lower mean command activity. A distinct burst-priority representative also improves IAE in the larger held-out burst. These benefits have explicit operating limits and a common-frequency shift cost; they establish conditional value rather than performance ordering across all delivery conditions.

The contribution is an explicit nonlinear bittide controller with both a rigorous reference-model margin result and independent empirical evidence of useful synchronization and resource tradeoffs. Extending the certificate to asynchronous delayed execution and testing broader networks and physical links are the next steps towards deployment under application-specific timing requirements.

Acknowledgments

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